



SD Specifications

Part 1

Standard Size SD Card Mechanical Addendum

**Version 4.00
June 15, 2011**

Addendum to:

**SD Specifications
Part 1 Physical Layer Specification
Version 4.00 May 30, 2011**

**Technical Committee
SD Card Association**

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Revision History

Date	Version	Changes compared to previous issue
March 27, 2008	1.00	The first release Mechanical specifications described in the Physical Layer Specification and Application Notes are merged in this Addendum.
	2.00	Skip this version
February 18, 2010	3.00	Add a detailed drawing to following sections: "3.2.Mechanical Form Factor and Dimensions." "5.3.miniSD to SD Card Adapter Mechanical Dimensions." "6.3.microSD to SD Card Adapter Mechanical Dimensions." Section 3.6 Thermal Specification for UHS-I Card is added.
June 15, 2011	4.00	Defines Standard Size SD UHS-II Card mechanical specification

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Conventions Used in This Document

Naming Conventions

- Some terms are capitalized to distinguish their definition from their common English meaning. Words not capitalized have their common English meaning.

Numbers and Number Bases

- Hexadecimal numbers are written with a lower case "h" suffix, e.g., FFFFh and 80h.
- Binary numbers are written with a lower case "b" suffix (e.g., 10b).
- Binary numbers larger than four digits are written with a space dividing each group of four digits, as in 1000 0101 0010b.
- All other numbers are decimal.

Key Words

- May: Indicates flexibility of choice with no implied recommendation or requirement.
- Shall: Indicates a mandatory requirement. Designers shall implement such mandatory requirements to ensure interchangeability and to claim conformance with the specification.
- Should: Indicates a strong recommendation but not a mandatory requirement. Designers should give strong consideration to such recommendations, but there is still a choice in implementation.

Application Notes

Some sections of this document provide guidance to the host implementers as follows:

Application Note: This is an example of an application note.

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1. General Description

1.1 Scope

This Addendum Specification defines mechanical specification of standard size SD Memory Card based on the Physical Layer Specification.

This document also contains Connector Specifications for SD Host system and SD adapters for miniSD card or microSD Card.

The usage of those notes shall be treated as mandatory requirements. It is always the product vender's (Card, Host, etc.) responsibility to design their products complied with the SD Specifications. At the same time, SD Card Association recommends the product vendors (Card, Host, etc.) designing products in conjunction with the Supplemental Note and Implementation Guideline.

1.2 Concept

This chapter describes the mechanical and electrical features, as well as the minimal recommendations for the SD Memory Card connector. The electrical features are applied to all types of SD Memory Cards and the mechanical features are applied to the Standard Size SD Memory Card. All technical drafts follow the DIN ISO standard.

The functions of the card package are:

- Protecting the chip
- Easy handling for the end user
- Reliable electrical interconnection
- Reliable write protect/card detection capability
- Bearing textual information and image
- Appealing appearance

The functions of the connector are:

- Attaching and fixing the card
- Electrical interconnecting the card to the system board
- Write protect/card detect indication
- Optional: switch on/off power supply
- Protection against card inverse insertion

The functions of the SD Card Adapters are:

- Providing the ability to use the microSD in a Standard Size SD Card socket.
- Providing the ability to use the miniSD in a Standard Size SD Card socket.

1.3 UHS-II Card Definition

This Version defines UHS-II Card Mechanical Specification (Chapter 2 and Chapter 3) but UHS-II Connector Specification (Chapter 4) will be defined in later versions.

2. Pin Assignment

2.1 Pin Assignment of Standard Size SD Card (Non UHS-II)

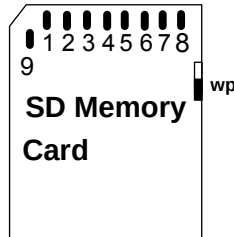


Figure 2-1: SD Memory Card Shape and Interface (Top View)

Figure 2-1 shows the general shape of the shape and interface contacts of the SD Memory Card. Table 2-1 defines the card contacts:

Pin #	SD Mode			SPI Mode		
	Name	Type ¹	Description	Name	Type ¹	Description
1	CD/DAT3 ²	I/O/PP ³	Card Detect/ Data Line [Bit 3]	CS	I ³	Chip Select (neg true)
2	CMD	PP	Command/Response	DI	I	Data In
3	V _{SS1}	S	Supply voltage ground	VSS	S	Supply voltage ground
4	V _{DD}	S	Supply voltage	VDD	S	Supply voltage
5	CLK	I	Clock	SCLK	I	Clock
6	V _{SS2}	S	Supply voltage ground	VSS2	S	Supply voltage ground
7	DAT0	I/O/PP	Data Line [Bit 0]	DO	O/PP	Data Out
8	DAT1 ⁴	I/O/PP	Data Line [Bit 1]	RSV		
9	DAT2 ⁵	I/O/PP	Data Line [Bit 2]	RSV		

1) S: power supply; I: input; O: output using push-pull drivers; PP: I/O using push-pull drivers;

2) The extended DAT lines (DAT1-DAT3) are input on power up. They start to operate as DAT lines after SET_BUS_WIDTH command. The Host shall keep its own DAT1-DAT3 lines in input mode, as well, while they are not used.

3) At power up this line has a 50KOhm pull up enabled in the card. This resistor serves two functions Card detection and Mode Selection. For Mode Selection, the host can drive the line high or let it be pulled high to select SD mode. If the host wants to select SPI mode it should drive the line low. For Card detection, the host detects that the line is pulled high. This pull-up should be disconnected by the user, during regular data transfer, with SET_CLR_CARD_DETECT (ACMD42) command

4) DAT1 line may be used as Interrupt Output (from the Card) in SDIO mode during all the times that it is not in use for data transfer operations (refer to "SDIO Card Specification" for further details).

5) DAT2 line may be used as Read Wait signal in SDIO mode (refer to "SDIO Card Specification" for further details).

Table 2-1: SD Memory Card Pad Assignment

2.2 Pin Assignment of Standard Size UHS-II SD Card

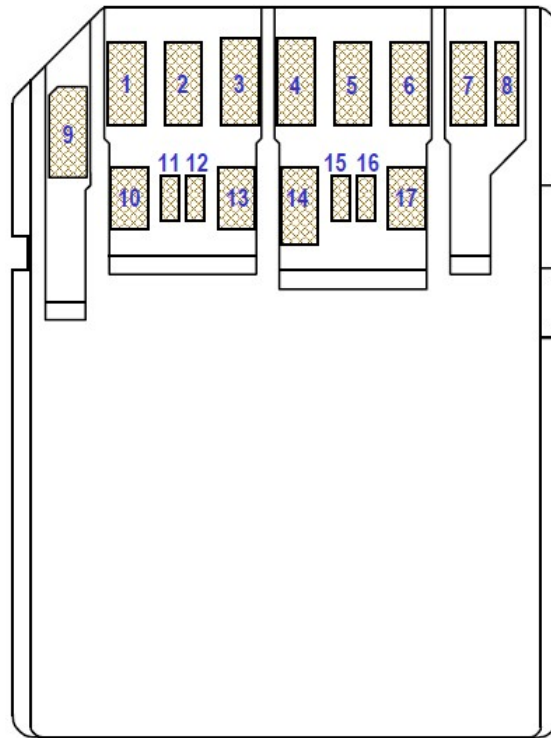


Figure 2-2 : Standard Size Card Shape and Interface for UHS-II

Pin#	SD Mode			UHS-II Mode		
	Name	Type	Description	Name	Type	Description
1	CD/DAT3	I/O/PP	Card Detect/ Data Line [Bit 3]	-		Not Used
2	CMD	PP	Command/Response	-		Not Used
3	VSS1	S	Supply voltage ground	VSS1	S	Supply voltage ground
4	VDD	S	Supply voltage	VDD1	S	Supply voltage (3.3V)
5	CLK	I	Clock	-		Not Used
6	VSS2	S	Supply voltage ground	VSS2	S	Supply voltage ground
7	DAT0	I/O/PP	Data Line [Bit 0]	RCLK+	I	
8	DAT1	I/O/PP	Data Line [Bit 1]	RCLK-	I	
9	DAT3	I/O/PP	Data Line [Bit 2]	-		
10	-		Not used (Connected to ground)	VSS3	S	Supply voltage ground
11	-		Not used	D0+	LVDS	
12	-		Not used	D0-	LVDS	
13	-		Not used (Connected to ground)	VSS4	S	Supply voltage ground
14	-		Not used	VDD2	S	Supply voltage (1.8V)
15	-		Not used	D1-	LVDS	
16	-		Not used	D1+	LVDS	
17	-		Not used (Connected to ground)	VSS5	S	Supply voltage ground

Table 2-2 : Standard Size UHS-II Card Pin Assignment

3. Mechanical Specification for Standard SD Memory Card

3.1 Card Package

Every card package shall have the characteristics described in this section.

3.1.1 Design and Format

Dimensions SD Memory Card package	24 mm x 32 mm; (min. 23.9 mm x 31.9 mm; max. 24.1 mm x 32.1 mm) Other dimensions: Refer to Figure 3-1 through Figure 3-4. Refer to Figure 3-1 through Figure 3-4 (for UHS-II and Non UHS-II) and Refer to Figure 3-10 through Figure 3-13 for UHS-II Card Testing according to MIL STD 883, Meth 2016
Thickness	'Inter Connect Area': 2.1 mm +/- 0.15 mm 1.4 mm +/- 0.15 mm for Thin SD Card. 'Substrate Area': Max 2.25 mm Max 1.55 for Thin SD Card Refer to Figure 3-2, Figure 3-6 (Thin SD) or Figure 3-11 (UHS-II)
Label or printable area	In 'Substrate Area' only Refer to Figure 3-3 (Non UHS-II) or Figure 3-13 (UHS-II).
Surface	Plain (except contact area)
Edges	Smooth edges, refer to Figure 3-2 and Figure 3-3 (Including UHS-II).
Inverse insertion	Protection on left corner (top view). Refer to Figure 4-4.
Position of ESC	Along middle of shorter edge. Refer to Figure 3-1 and Figure 3-2 for Non UHS-II Card. Refer to Figure 3-10 through Figure 3-12 for UHS-II Card

Table 3-1 : SD Memory Card Package – Dimensions

3.1.2 Reliability and Durability

Card Ambient Temperature (T _a) (Note 1)	Operation Temperature: -25 to 85 deg.C Storage Temperature : -40 to 85 deg.C Storage Temperature test condition: -40 deg.C (168h)/85 deg.C (500 h)
Moisture and Corrosion	Operation: 25 deg.C /95% relative humidity Storage: 40 deg.C /93% relative humidity Storage Temperature test condition: 40 deg.C /93% relative humidity/500h Salt water spray: 3% NaCl/35C; 24 h acc. MIL STD Method 1009
Durability	10.000 mating cycles; Test procedure: tbd.
Bending	10 N (Note 2: Appendix E.1)
Torque	0.15Nm, +/-2.5 deg. max. (Note 2: Appendix E.2)
Drop Test	1.5 m free fall
UV Light Exposure	UV: 254 nm, 15 Ws/cm ² according to ISO 7816-1
X-ray Exposure	0.1 Gy of medium-energy radiation (70 keV to 140 keV, cumulative dose per year) to both sides of the card, according to ISO7816-1.
Visual Inspection Shape and Form	No mold skin; complete form; no cavities surface smoothness <= -0.1 mm/cm ² within contour; no cracks; no pollution (fat, oil dust, etc.) (Note 2: Appendix E.3 and E.4)

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Minimum Moving Force of WP Switch	0.4N min. at any point moving 0mm to 0.4mm from the edge of the card case. (Ensures that the WP switch will not slide while it is inserted to the connector). (Note 2: Appendix E.5)
WP Switch Cycles	Minimum 1000 Cycles (@ Slide force 0.4 N to 5 N)
Rib Strength	Tolerance of 4N force 1 second. (Note 3)

Note 1: Refer to Section 3.6 for the thermal specification of UHS-I Card

Note 2: The test methods are shown in Appendix E.

Note 3: The metal pillar with 1.5mm diameter is stood at right angles to a contact pad. From the side of the middle of each rib, each rib is pushed and applied the force of 4N for 1 second by this pillar. By this pressurization, each rib shall not be broken and junction shall not be separated. Moreover, after 10 minutes from removing pressurization, each rib transformation shall not overlap the minimum width of contact pads (0.9mm-1.4mm).

Table 3-2 : Reliability and Durability

Note: Connector Durability, Card Thermal Conditions, Card Mechanical Performance Torque, Vend, etc for UHS-II Card will be revised in next version.

3.1.3 Electrical Static Discharge (ESD) Requirements

Refer to the Physical Layer Version 3.01 or later about ESD requirements.
The ESD test methods are shown in Appendix D.

3.1.4 External Signal Contacts (ESC)

	Non UHS-II Card	UHS-II Card
Number of ESC	9	17
Distance from front edge	1.2 mm	Refer to Figure 3-10 through Figure 3-12
ESC grid	2.5 mm	Refer to Figure 3-10 through Figure 3-12
Contact dimensions	1.7 mm x 4.0 mm	Refer to Figure 3-10 through Figure 3-12
Electrical resistance	30m-ohm (worst case: 100m-ohm)	30m-ohm (worst case: 100m-ohm)

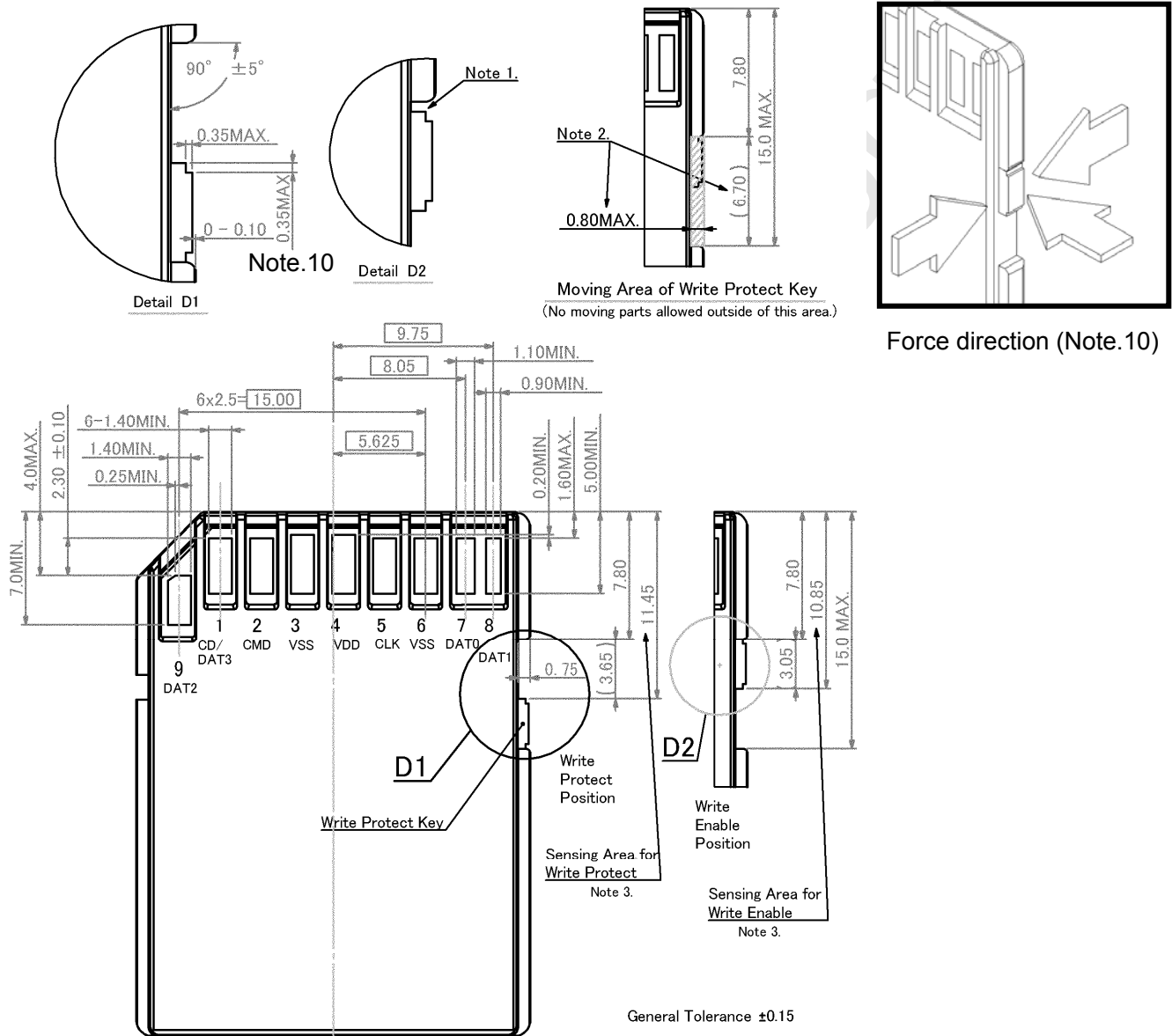
Table 3-3 : Standard Size SD Package - External Signal Contacts

3.1.5 Discontinuity and Micro-Interrupt

Refer to Section 4.1.3 Vibration and Shock about discontinuity and micro-interrupt when the card is inserted to a connector.

3.2 Mechanical Form Factor and Dimensions

The thickness of SD Memory Card package is defined as 2.1+/-0.15 mm. Refer to Figure 3-1 through Figure 3-4.



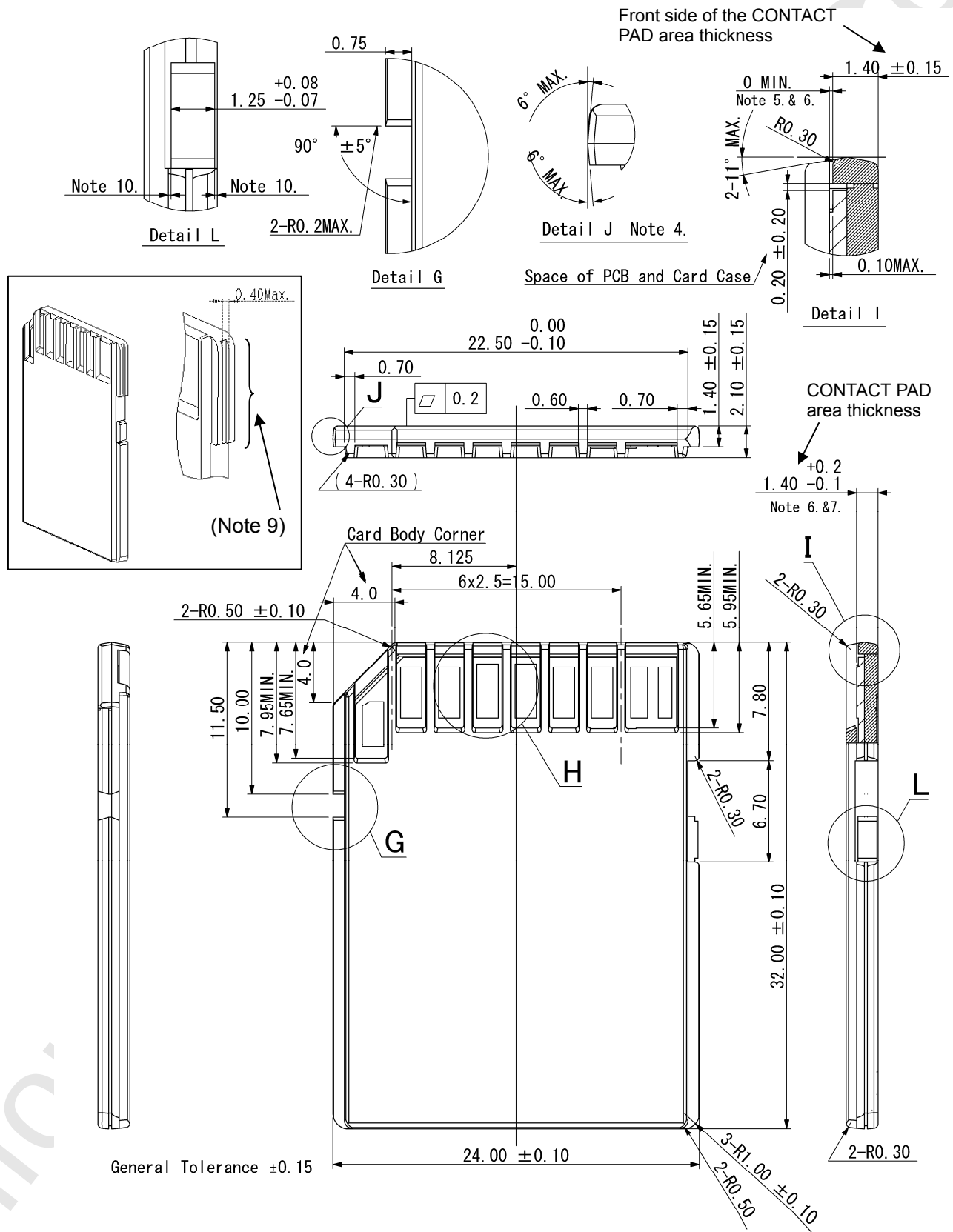
Note 1: Refer to Table 3-2 about minimum moving force of WP-key.

Note 2: The card width of the rest of this hatched area should be 24.0 ± 0.1 .

Note 3: The figure shown on the drawing is the dimension when the key is pressed towards the direction of each position.

Figure 3-1 : SD Memory Card - Mechanical Description (1 out of 4)

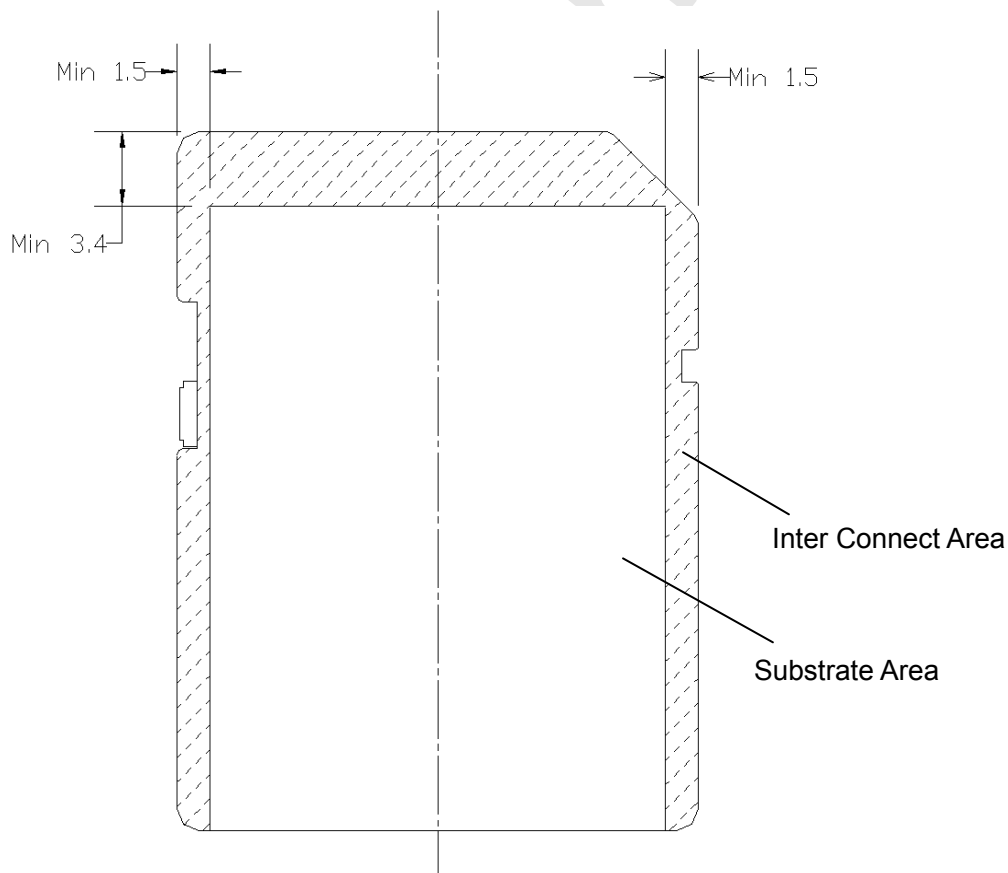
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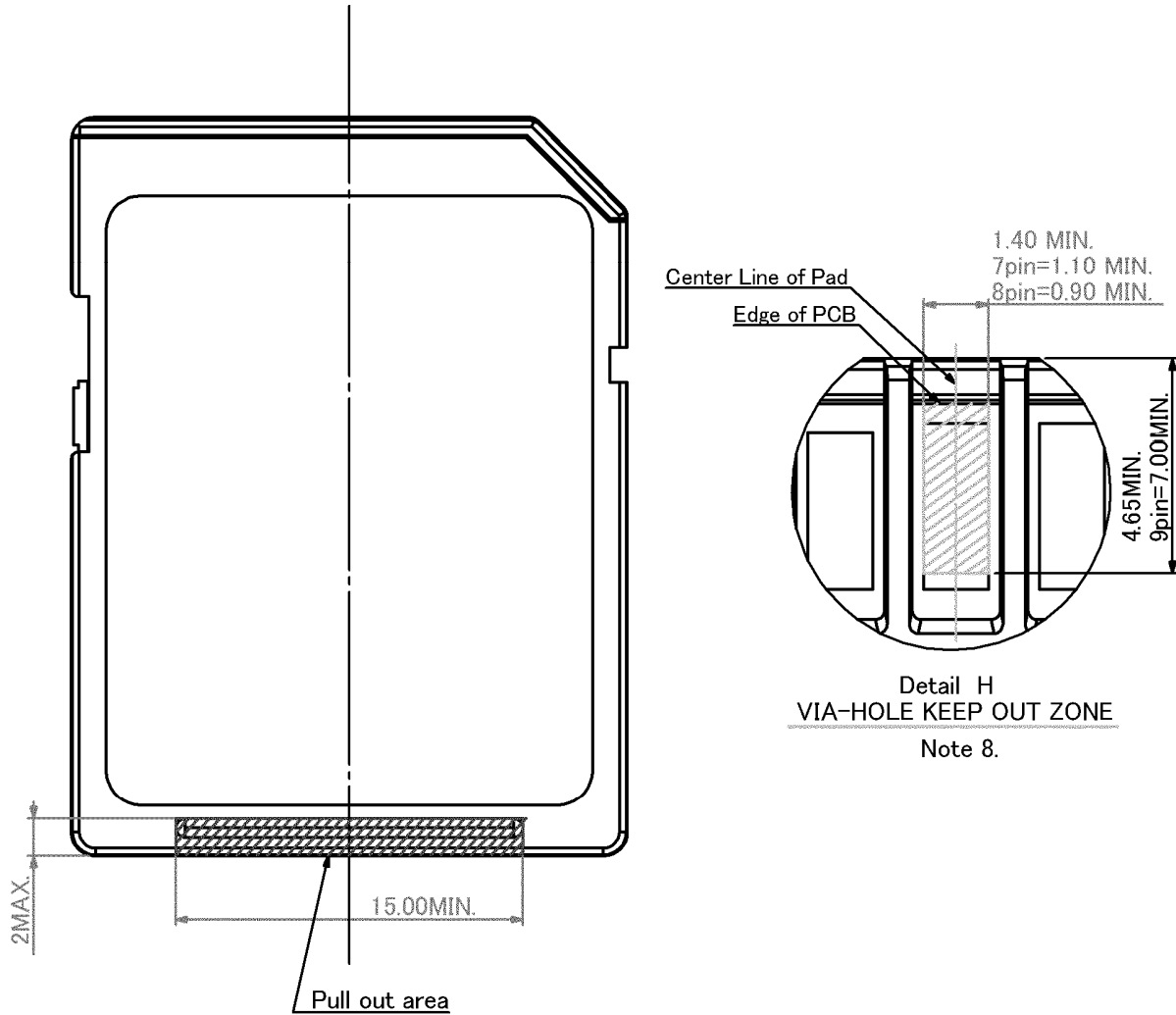
- Note 4: This detail figure specification shall be applied for the 3-sides of the card excluding detail I.
- Note 5: This dimension is from the surface of the contact pad to the card case.
 (1) The CONTACT PAD (GOLD PLATE) area thickness is $1.4+0.2\text{mm}$, $1.4-0.1\text{mm}$.
 (2) The front side of the CONTACT PAD area thickness is $1.4+0.15\text{mm}$, $1.4-0.15\text{mm}$.
 The card shall satisfy (1) and (2). In addition, "0 Min" indicates that the thickness of the CONTACT PAD area shall be larger than that of the front side of the CONTACT PAD area so that the card can be pushed out smoothly.
- Note 6: These dimensions shall be satisfied when no force is put on any pad (0 N) and also when a 0.5N force is applied to the measure point.
- Note 7: This dimension is from the surface of the contact pad to the surface of the label side of the card case.
- Note 9: The gap dimension of side is indicated in the drawing.
- Note 10: The write protect key may not extrude from the card surface. The key shall retreat inside of the surface with less than 0.3N force, if the key extrudes from the surface.

Figure 3-2 : SD Memory Card - Mechanical Description (2 out of 4)



Note: Refer to Table 3-1 about Thickness and Label or Printable Area.

Figure 3-3 : SD Memory Card - Mechanical Description (3 out of 4)



Note 8: The via-hole should not be positioned within hatched area.

Figure 3-4 : SD Memory Card - Mechanical Description (4 out of 4)

Note that even though the WP switch appears in the given diagram, it was defined as an optional for a Thin Standard Size SD Memory Card.



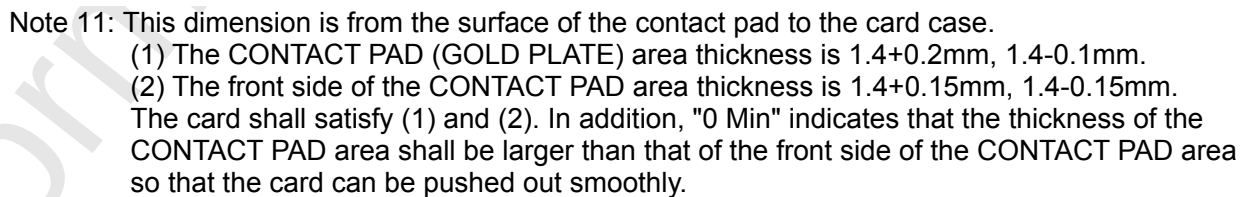


Figure 3-6 : Mechanical Drawing of Thin SD Memory Card (2 out of 2)

3.4 Nonconductive Area

Figure 3-7 shows "Nonconductive Areas" in Standard Size SD Memory Card. Conductive materials shall not be included in the hatching areas. Conductive area (except for pads) should be connected to Vss.

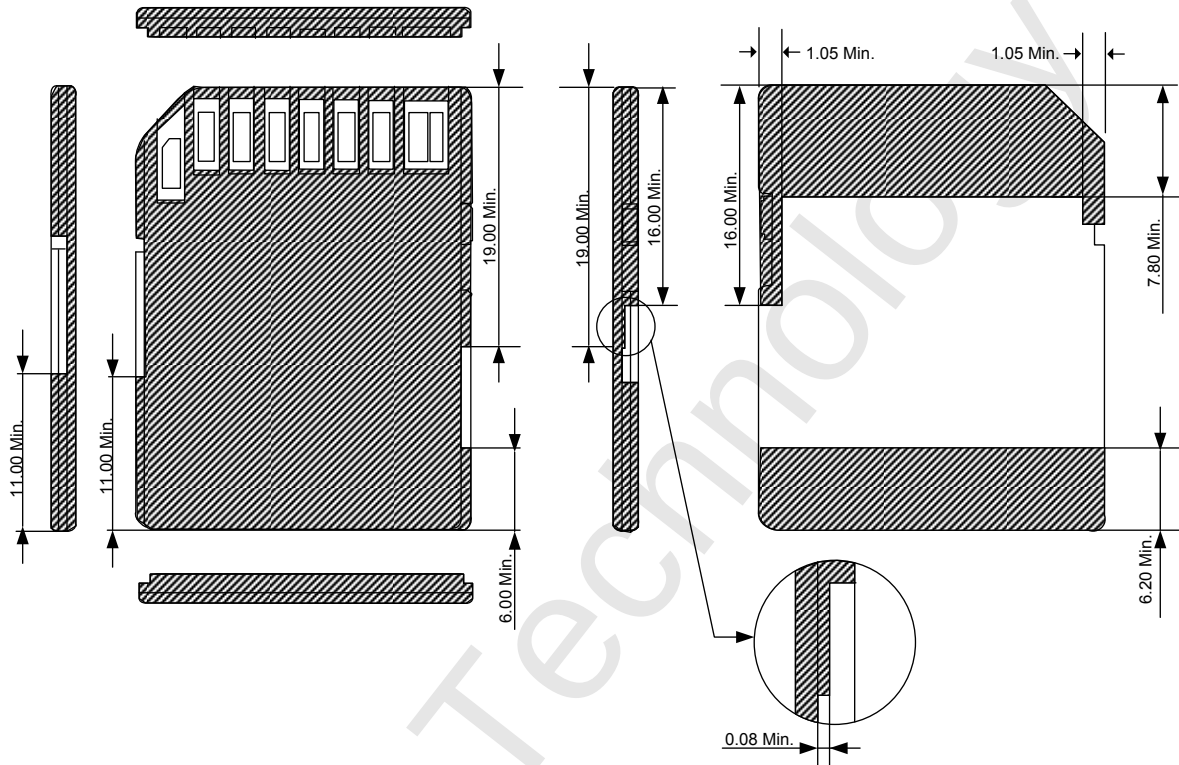


Figure 3-7 : Nonconductive Areas of SD Memory Card

Following notes are applicable for all the mechanical descriptions from Figure 3-1 through Figure 3-7 (including Thin SD Memory Card):

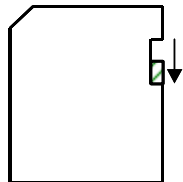
- (1): The numbers enclosed by a square indicate the distance between base lines. Those values are for information only (the given general tolerances are not related to them).
- (2): Pins #3 and #4 are located "0.2 Min." ahead of pins #1, #2, #5, #6, #7 and #8.
- (3): "1.6 Max." is for pins #1, #2, #5, #6, #7, #8.

3.5 Write Protect switch position

Figure 3-8 describes the Write Protect switch position for all cases and card types.

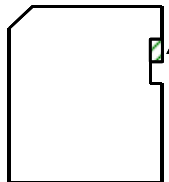
-R/W card(WP switch is movable)

Write unable



Pulling down
the WP Switch.

Write enable

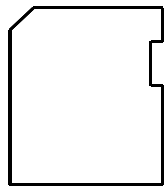


Pulling up the
WP Switch.



This is the case of 1.4mm
card which does not support
the WP switch.

-ROM card(WP area is fixed)



This form is same as
taking off the WP Switch
from R/W card.

Figure 3-8 : WP Switch Definition for All Cases and Card Types

3.6 Thermal Specification for UHS-I Card

The thermal specification in Table 3-2 is defined with the condition of current consumption up to 200mA implicitly (the maximum current in High Speed mode). The temperature requirement is specified by the Ambient Temperature (T_a), presuming that heat dissipated by a card is to ambient air around a card without air blow. In this case, Ambient Temperature should be interpreted as internal temperature of a host device.

The UHS-I card has several operating modes with different current consumption. Host system needs to support another effective heat removal method when UHS-I card is used in higher performance operating mode more than 200mA. For the system supporting a new heat removal method, the thermal specification should be defined by "**Card Case Temperature (T_c)**", in another word, a card surface temperature. Heat removal performance of a host can be calculated by using thermal resistance between various surfaces in the system and card case temperature T_c .

Figure 3-9 shows an example heat removal path to use UHS-I card. T_{ai} stands for Ambient Temperature of internal to host device. T_{ae} stands for Ambient Temperature of external to host device. The heat removal method shown in this figure is based on heat transfer through materials of host - heat dissipated by a card is to external ambient air through connector, PCB and body of host. If T_c is kept under the defined value, T_{ai} can be more than T_c up to 85 deg.C. The characteristics of a connector have a great influence on heat removal performance of the system. The minimum operating temperature is $T_{ai} > -25$ deg.C. The range of T_{ae} depends on host specification.

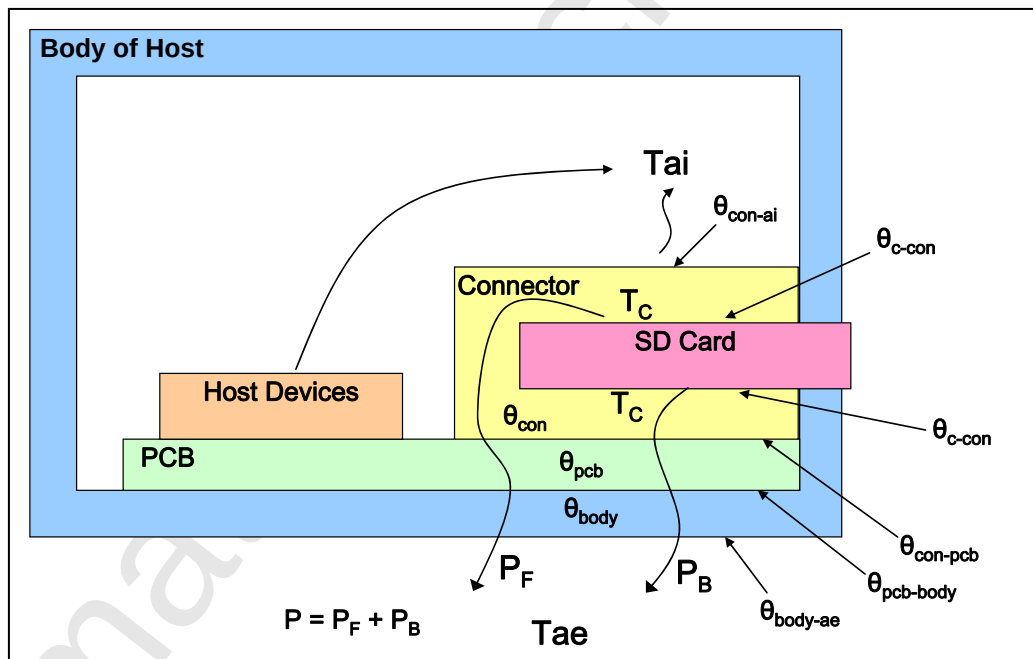


Figure 3-9 : Example Heat Removal Path for UHS-I

θ_{con} :	Thermal Resistance of a Connector
θ_{pcb} :	Thermal Resistance of a PCB
θ_{body} :	Thermal Resistance of a Host Body
θ_{con-ai} :	Thermal resistance between a connector and ambient air internal to a host
$\theta_{body-ae}$:	Thermal resistance between host body and ambient air external to a host
$\theta_{con-pcb}$:	Thermal resistance between a connector and PCB
θ_{c-con} :	Thermal resistance between card case and connector

$\theta_{pcb-body}$:	Thermal resistance between PCB and host body
T_{ai} :	Temperature of Ambient Air internal to host device
T_{ae} :	Temperature of Ambient Air external to host device
T_C :	Maximum Card Case Temperature
P_F :	Power consumption dissipated from front side
P_B :	Power consumption dissipated from back side
P :	Total power consumption $P = P_F + P_B$

3.6.1 Thermal Specification Based on Ambient Temperature

The host system supporting the heat removal method based on Ambient Temperature can use any card in Default Speed mode or High Speed mode (3.3V signaling). When using UHS-I card in one of UHS-I modes (1.8V signaling), host shall select an operating mode up to 200mA (SDR12, SDR25 or using current limit function). High performance feature is not available in this host system even if UHS-I card is used. The thermal specification based on Ambient Temperature is defined by Table 3-2.

3.6.2 Thermal Specification Based on Card Case Temperature

The host system supporting the heat removal method based on Card Case Temperature (T_C) can use UHS-I card in a higher performance operating mode. The maximum current a host can supply depends on the level of heat removal method taken by the host and power supply capability to a card. The card current consumption can be controlled by Current Limit function in CMD6 Function Group 4. So host system needs to set an appropriate current limit value to use SDR50, SDR104 or DDR50.

By defining T_C , thermal design of card and host can be separated. Heat generated in a card is removed through the card surface. In case of thin card (whose front and back surface area is much larger than surface area of sides), the heat is considered to be mainly removed through two surfaces: front and back (contact pads may also play the role of removing some). Host system needs to implement a heat removal method to maintain Card Case Temperature on front and back surfaces. Forced convection, where heat is removed by air flow or the use of heat fins are possible ways of maintaining to T_C .

The maximum Card Case Temperatures are defined by Table 3-4 depends on card current consumption of UHS-I modes. The minimum operating temperature is defined by ambient temperature ($T_{ai} > -25$ deg.C).

Card Current Consumption [mA]	200	400	600	800
Card Case Temperature [deg.C]	87	80	TBD	TBD

Table 3-4 : Maximum Card Case Temperature

Note: The maximum current is defined as 500mA in the Physical Layer Specification Version 2.00 and then most existing connectors are designed to be able to use up to 500mA. As further consideration is required to support more than 400mA, Card Case Temperatures for 600mA and 800mA are TBD in this version of mechanical Addendum and will be defined in the later version.

3.6.3 Storage Temperature

Storage temperature is defined by Table 3-2 for all types of cards including UHS-I cards. Long term storage without operation is presumed to be at storage temperature.

3.6.4 UHS-II Card

Basically the UHS-I Card thermal specification is also applied to UHS-II Card but this section will be revised for UHS-II Card in next version.

3.7 Mechanical Specification for UHS-II SD Card

The following 4 technical figures and 3 dimension tables define the card package of UHS-II SD Memory Card with 2.1mm card thickness.

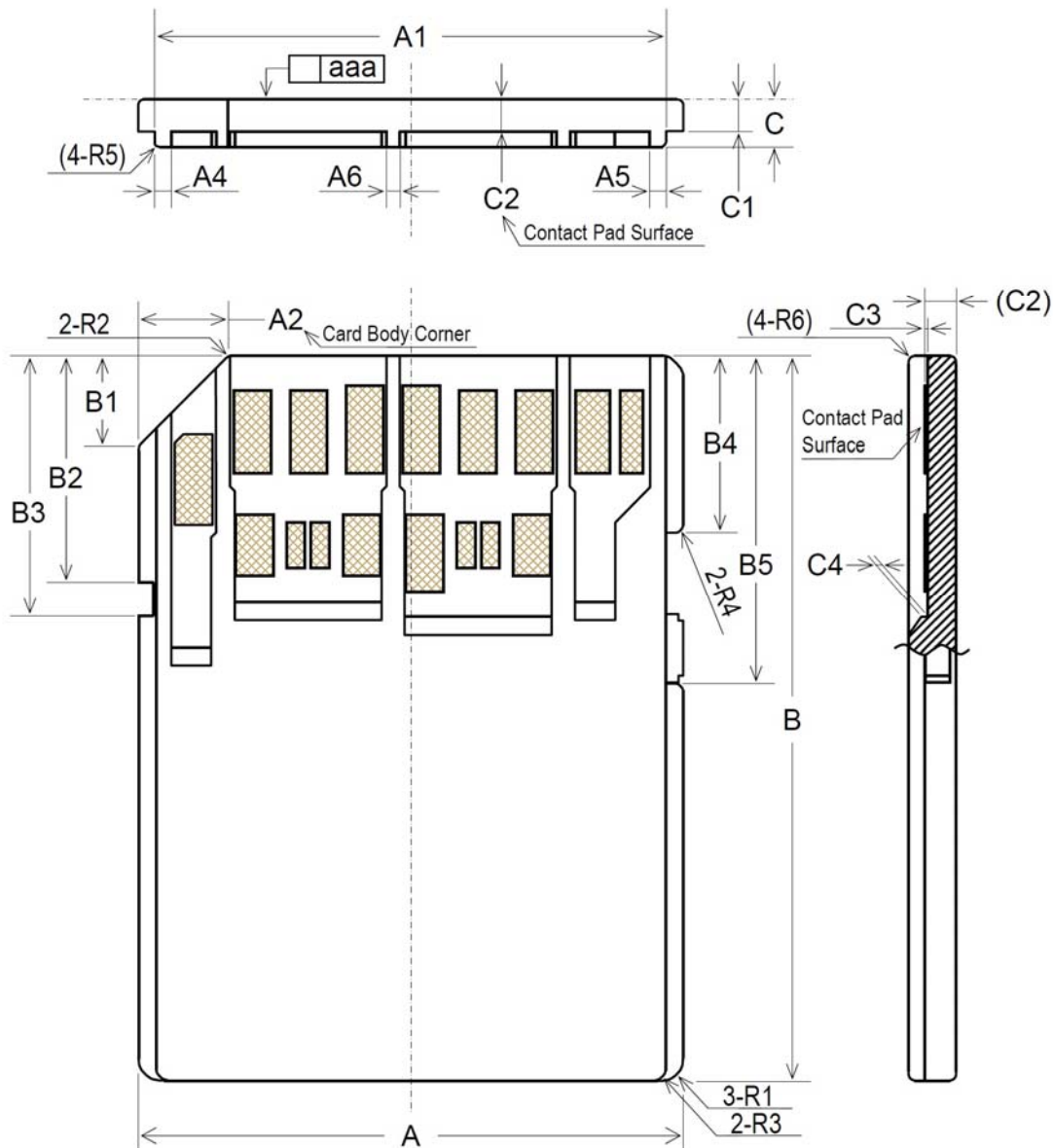


Figure 3-10 : Standard Size UHS-II SD Memory Card - Over View (1 out of 4)

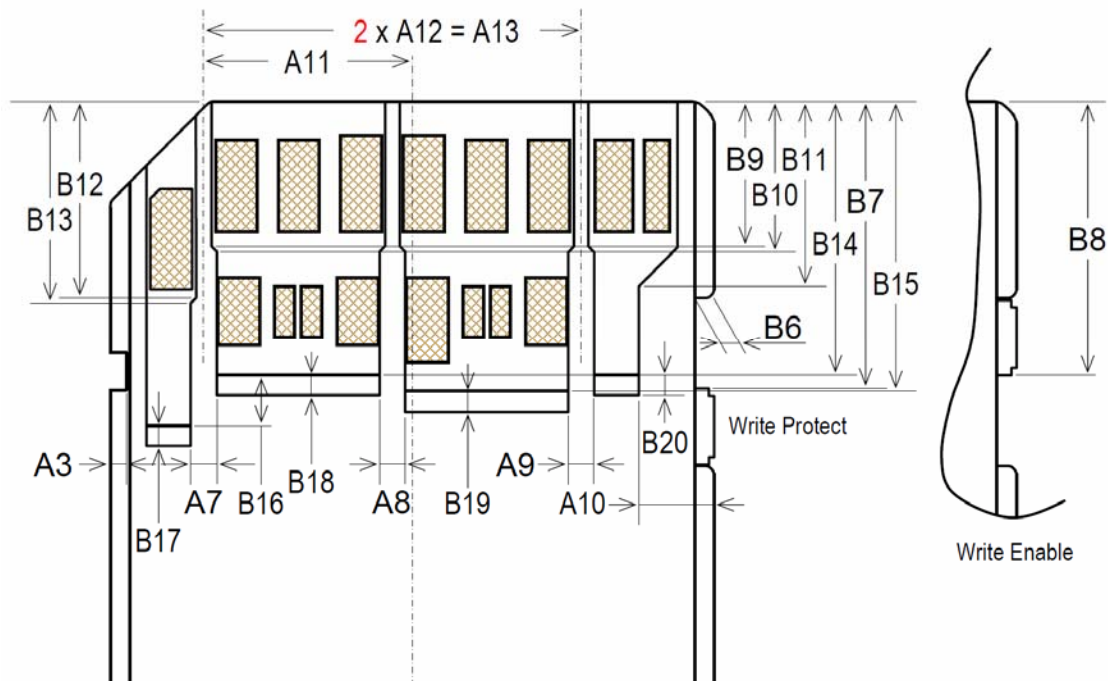


Figure 3-11 : Standard Size UHS-II SD Memory Card – Periphery of Contact Pads (2 out of 4)

B17 to B20 indicate taper (slope) area.

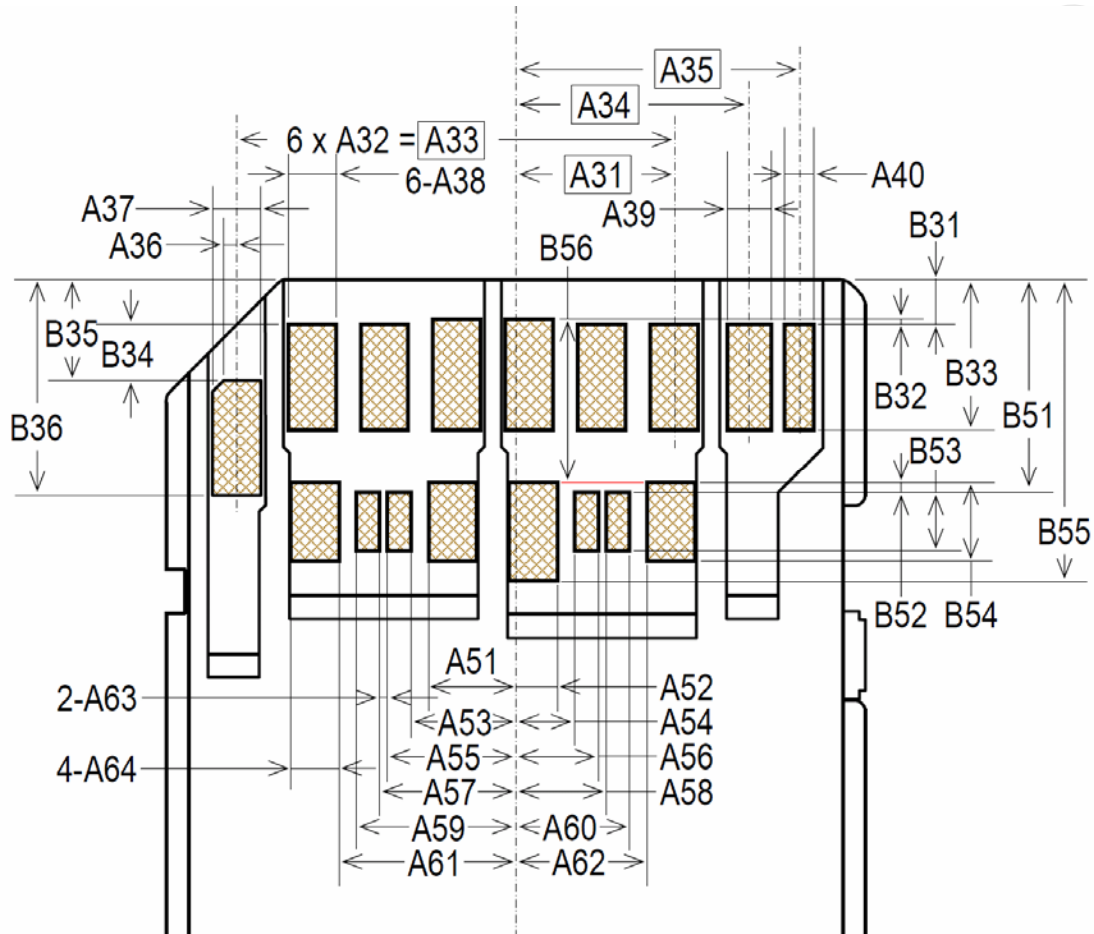


Figure 3-12 : Standard Size UHS-II SD Memory Card – Contact Pads (3 out of 4)

Non UHS-II Card has ribs between all contact pads. However, the ribs of UHS-II card are thinned down as described above.

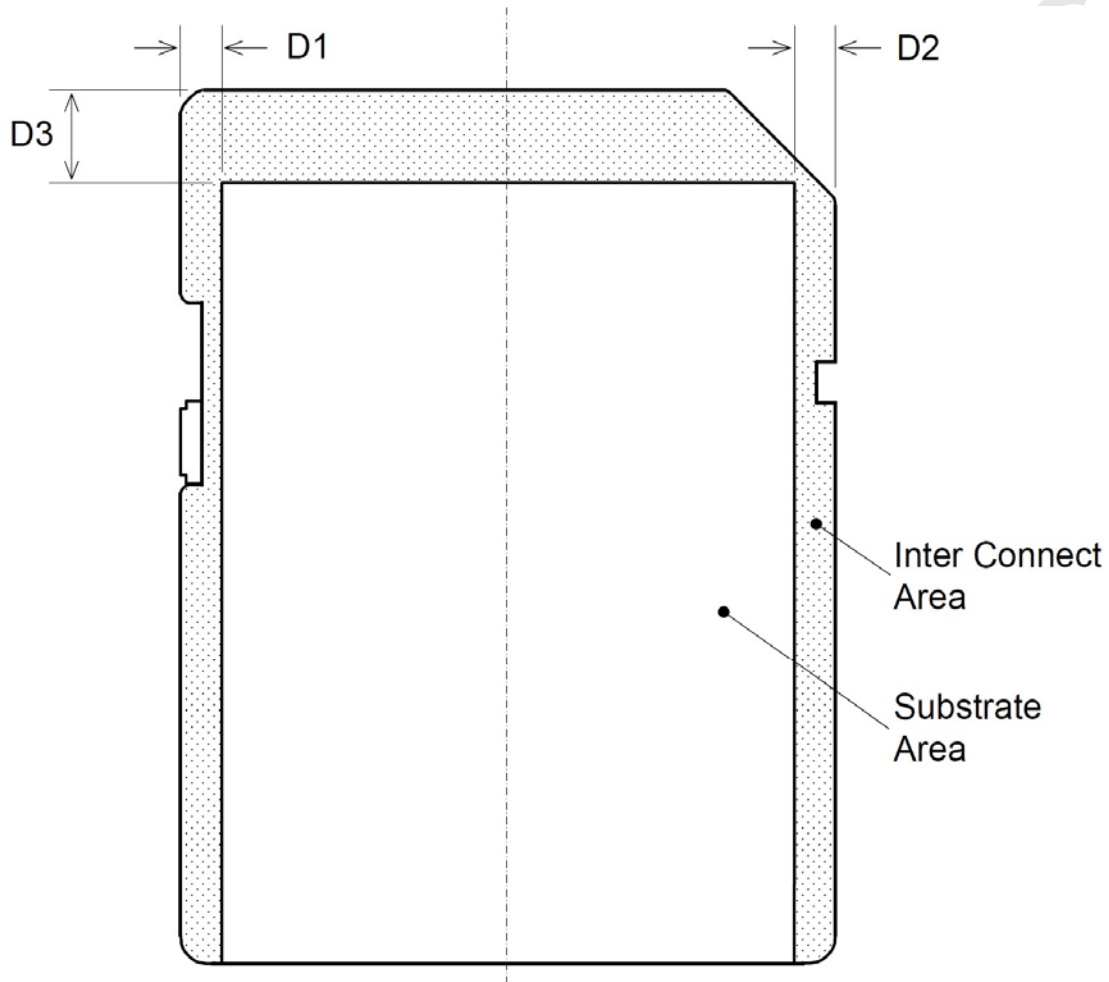


Figure 3-13 : Standard Size UHS-II SD Memory Card - Bottom View (4 out of 4)

Labeling and printable area shall be within Substrate Area. Then it is recommended that Connector should use Inter Connect Area as card support area.

Standard Size SD Card Mechanical Addendum Version 4.00

SYMBOL	COMMON DIMENSIONS			NOTE
	MIN	NOM	MAX	
A	23.9	24.0	24.1	
A1	22.4	22.5	22.5	
A2	3.85	4.00	4.15	
A3	0.60	0.75	0.90	
A4	0.55	0.70	0.85	
A5	0.55	0.70	0.85	
A6	0.45	0.60	0.75	
A7	0.90	1.05	1.20	
A8	0.90	1.05	1.20	
A9	0.90	1.05	1.20	
A10	2.85	3.00	3.15	
A11	7.975	8.125	8.275	
A12	-	7.50	-	
A13	14.85	15.00	15.15	
A31	-	5.625	-	BASIC
A32	-	2.5	-	
A33	-	15.00	-	BASIC
A34	-	8.05	-	BASIC
A35	-	9.75	-	BASIC
A36	0.25	-	-	
A37	1.40	-	-	
A38	1.40	-	-	
A39	1.10	-	-	
A40	0.90	-	-	
A51	2.675	2.825	2.975	
A52	1.425	1.575	1.725	
A53	3.325	3.475	3.625	
A54	2.075	2.225	2.375	
A55	4.125	4.275	4.425	
A56	2.875	3.025	3.175	
A57	4.325	4.475	4.625	
A58	3.075	3.225	3.375	
A59	5.125	5.275	5.425	
A60	3.875	4.025	4.175	
A61	5.775	5.925	6.075	
A62	4.525	4.675	4.825	
A63	0.20	-	-	*9
A64	1.65	-	1.85	

1. Dimensions are in millimeters.
2. 2nd row GND pads should be connected with 1st row GND pad inside the card.
3. VDD2 pad must not be connected with the contact pins for xD of a combo connector.
4. Whichever of VDD1 and VDD2 may be connected first.
5. However, the legacy signal pins should be connected after VDD1.
6. The UHS-II signal pins should be connected after VDD2.
7. When a card is pushed in front of the lock position, the signal and GND contact pins are allowed to disconnect once.
8. Also refer to detailed description of Standard Size SD Card Mechanical Specification Version 3.00 for the legacy shape portion.
9. This is a double dimension. It shall satisfy this rule, after satisfying other rules.

Table 3-5 : Dimensions Table for Standard Size UHS-II SD Memory Card (1 out of 3)

Standard Size SD Card Mechanical Addendum Version 4.00

SYMBOL	COMMON DIMENSIONS			NOTE
	MIN	NOM	MAX	
B	31.90	32.00	32.10	
B1	3.85	4.00	4.15	
B2	9.85	10.00	10.15	
B3	11.35	11.50	11.65	
B4	7.65	7.80	7.95	
B5	14.35	14.50	14.65	
B6	0.60	0.75	0.90	
B7	11.30	11.45	11.60	
B8	10.70	10.85	11.00	
B9	5.65	5.80	5.95	
B10	5.875	-	-	
B11	7.20	7.35	7.50	
B12	7.65	7.80	7.95	
B13	7.875	-	-	
B14	10.75	10.90	11.05	
B15	11.40	11.55	11.70	
B16	1.85	2.00	2.15	
B17	0.65	0.80	0.95	
B18	0.65	0.80	0.95	
B19	0.65	0.80	0.95	
B20	0.65	0.80	0.95	
B31	-	-	1.60	
B32	0.20	-	-	
B33	5.00	5.15	5.30	
B34	2.20	2.30	2.40	
B35	-	-	4.00	
B36	7.00	7.15	7.30	
B51	7.20	7.35	7.50	
B52	0.275	0.35	0.425	
B53	1.925	2.00	2.075	
B54	2.625	2.70	2.775	
B55	10.25	10.40	10.55	
B56	5.65	-	-	*9
C	1.95	2.10	2.25	
C1	1.25	1.40	1.55	
C2	1.30	1.40	1.60	
C3	0.00	-	-	
C4	-	-	0.30	

Table 3-6 : Dimensions Table for Standard Size UHS-II SD Memory Card (2 out of 3)

Standard Size SD Card Mechanical Addendum Version 4.00

SYMBOL	COMMON DIMENSIONS			NOTE
	MIN	NOM	MAX	
R1	0.90	1.00	1.10	
R2	0.40	0.50	0.60	
R3	0.35	0.50	0.65	
R4	0.15	0.3	0.45	
R5	0.15	0.3	0.45	
R6	0.15	0.3	0.45	
D1	1.50	-	-	
D2	1.50	-	-	
D3	3.4	-	-	
aaa	-	-	0.20	

Table 3-7 : Dimensions Table for Standard Size UHS-II SD Memory Card (3 out of 3)

4. Connector

There are 9 contact springs for contacting the first row (all SD connectors) and 8 contact springs for contacting second Row of UHS-II Card (only UHS-II connector). In case of UHS-II Connector, the connector scratches SD card first row pads twice when the card is inserted because contact springs of second row pass through the first row. Consequently, UHS-II connector will deteriorate Durability Cycles of insertion and removable. The detail of the durability of UHS-II interconnect system will be defined after observations. UHS-II connector specification will be defined in next version.

The connector housing which accommodates the card is formed of plastic. When the connector supports Standard Size SD Memory Card, a Write Protect/Card Detection switch shall be part of the housing in order to be able to detect the position of the Write Protect sliding tablet on the card. Testing procedures are performed according to DIN IEC 68.

For an SD Memory Card, the Write Protect switch is always mandatory for the card side. Note that for hosts that support Read Only operations, the Write Protect detection capability in the SD Connector (Host side) is not necessary.

4.1 SD Connector Reliability

The Standard Size SD Card Connector shall satisfy all reliability test requirements of this section. Unless otherwise specified, all test measurements shall be made at:

Temperature	15deg.C to 35deg.C
Air pressure	86 to 106 kPa
Relative humidity	25% to 85%

If above conditions are too moderate to obtain reproducible results, the following conditions should be used.

Temperature	23deg.C +/- 1deg.C
Air pressure	86 to 106 kPa
Relative humidity	50% +/- 2%

Note : You may use either a real SD Card or/and a dummy test card whether it complies with the SD Card specification. The cards may be supplied by a card vender (preferable) or self-built by a connector manufacturer.

Office Environment [cycles]	10,000 minimum
Harsh Environment [cycles]	5,000 minimum

Note : These numbers are defined for non UHS-II Standard Size SD connector.

4.1.1 SD Connector Mechanical Performance

This section specifies the mechanical performance of the Standard Size SD Card Connector.

Office Environment

STANDARD	TESTING
Guaranteed number of insertions/extractions = 10,000 minimum	Refer to Office Environment, Section 4.1.5

Harsh Environment

STANDARD	TESTING
Guaranteed number of insertions/extractions = 5,000 minimum	Refer to Harsh Environment, Section 4.1.5

Total Insertion Force

STANDARD	TESTING
40 N maximum	Insert at speed of 25 mm/minute Except influence of the force for eject and card lock mechanism

Total Pulling Force

STANDARD	TESTING
0.5N minimum and 40N maximum	Extract at speed of 25 mm/minute Except influence of the force for eject and card lock mechanism

Notes: Minimum total pulling force is relaxed from 1.0N to 0.5N because it is difficult to satisfy.

Voltage Proof

STANDARD	TESTING
a. No shorting or other damages when 500 Vrms AC is applied for 1 minute b. Current leakage 1 mA maximum	IEC 60512-4-1

Insulation Resistance

STANDARD	TESTING
a. Initially 1000 M-ohm minimum b. After test 100 M-ohm minimum	IEC 60512-3-1. Measure within 1 minute after applying 500V DC.

Temperature Rise

STANDARD	TESTING
0.5 A per contact	IEC 60512-5-1. Based upon 30deg.C rise above ambient temperature.

4.1.2 SD Connector Environmental Performance

Operating Environment

STANDARD
Operating Temperature: -25deg.C to +85deg.C Relative humidity: 95% maximum (non-condensing)

Storage Environment

STANDARD
Storage Temperature: -40deg.C to +85deg.C Relative humidity: 95% maximum (non-condensing)

Notes: Maximum temperature is changed from 90deg.C to +85deg.C in conformity with the miniSD / microSD connector specification).

4.1.3 SD Connector Environmental Resistance

Damp Heat, Cyclic

STANDARD	TESTING
Per Contact Resistance (low level) Section, Part b Per Insulation Resistance Section, Part b	IEC 60512-11-12 10 cycles (1 cycle = 24 hours) with connectors engaged

Rapid Change Of Temperature

STANDARD	TESTING
No physical damage shall occur during testing Per Contact Resistance (low level) Section, Part b Per Insulation Resistance Section, Part b	IEC 60512-11-4 -55deg.C to +85deg.C 5 minute transition time (max) 5 cycles (1 cycle = 1 hour) with connectors engaged

Dry Heat

STANDARD	TESTING
Per Contact Resistance (low level) Section, Part b	IEC 60512-11-9 85deg.C, 96 hours with connectors engaged. Exclude load and insulation resistance measurements

Vibration

STANDARD	TESTING
a. No mechanical damage shall occur on the parts b. Shall not cause current interruption greater than 100 ns	IEC 60512-6-4 20 m/s ² peak amplitude, 10 Hz to 2000 Hz, 5 minutes per 1 cycle, 10 cycle per 1 axis total 30 cycles per 3 axis.

Note: Method of measuring discontinuity associated with connector: At DC 5V and 150mA Max.; decent of voltage more than 50% (= less than 2.5V) as discontinuity. Discontinuity specification for the connector: Maximum 100nsec. of discontinuity period.

Shock

STANDARD	TESTING
a. No mechanical damage shall occur on the parts b. Shall not cause current interruption greater than 100 ns	IEC 60512-6-3 Acceleration 490 m/s ² Standard holding time 11 ms, semi-sine wave, velocity change 3.44m/s.

Note: Method of measuring discontinuity associated with connector: At DC 5V and 150mA Max.; decent of voltage more than 50% (= less than 2.5V) as discontinuity. Discontinuity specification for the connector: Maximum 100nsec. of discontinuity period.

4.1.4 SD Connector Electrical Performance

This section specifies electrical performance of the Standard Size SD Card Connector.

Contact Resistance-Millivolt Level Method

STANDARD	TESTING
a. Initially 100 milli-ohm maximum	IEC 60512-2-1 Open voltage 20 mV Test current 1 mA a. Measure and record the initial resistance (R_i) of the separate connector contact interface. Refer to Figure 4-1. R_i 100 milli-ohm
b. After test 40 milli-ohm maximum change	b. Measure and record the resistance after test (R_f) of the connector system. Resistance value after test: $R_f = R_i \pm 40$ milli-ohm

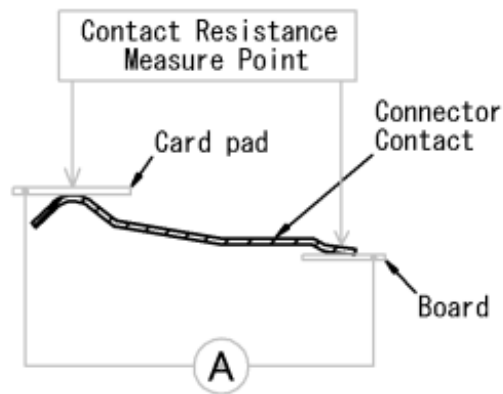


Figure 4-1 : Contact Resistance Measurement method

Cold

STANDARD	TESTING
Per Contact Resistance (low level) Section, Part b	IEC 60512-11-10 -25deg.C, 96 hours with connectors engaged

Damp Heat, Steady State

STANDARD	TESTING
Per Contact Resistance (low level) Section, Part b	IEC 60512-11-3 Steady State 40deg.C, 90 to 95% RH 96 hours with connectors engaged
Per Insulation Resistance Section, Part b	

Hydrogen Sulfide

STANDARD	TESTING
Per Contact Resistance (low level) Section, Part b	JEIDA 38 3 PPM hydrogen sulfide 40deg.C, approx. 80% RH 96 hours, with connectors engaged

4.1.5 SD Connector Environmental Durability

The Standard Size SD Card Connector shall meet below two environmental requirements.

Test conditions for the mate/unmate cycles are:

Cycle Rate 400-600 cycles per hour

Temperature 15deg.C to 35deg.C

Relative Humidity 25% to 85%

Air Pressure 86 to 106 kPa

Office Environment

The office environment is defined in EIA-364-B Class 1.1 - year round air conditioning (non-filtered) with humidity control.

Contact resistance - Part a
Mate and unmate the connector for a total of 10,000 cycles
Contact resistance - Part b

Table 4-1 : Office Environment Durability

Harsh Environment

The harsh environment is defined in EIA-364-B Class 1.3—no air conditioning, no humidity control with normal heating and ventilation:

Contact resistance - Part a	
Mate and unmate the connector 1,000 cycles	TOTAL CYCLES = 1,000
Damp heat, cyclic Section (1 cycle=24hours)	Total 1 cycle
Mate and unmate the connector 1,000 cycles	TOTAL CYCLES = 2,000
Damp heat, cyclic Section (1 cycle=24hours)	Total 2 cycle
Mate and unmate the connector 3,000 cycles	TOTAL CYCLES = 5,000
Damp heat, cyclic Section (1 cycle=24hours)	Total 3 cycle
Hydrogen sulfide per Hydrogen Sulfide Section	96 hours
Contact resistance - Part b	

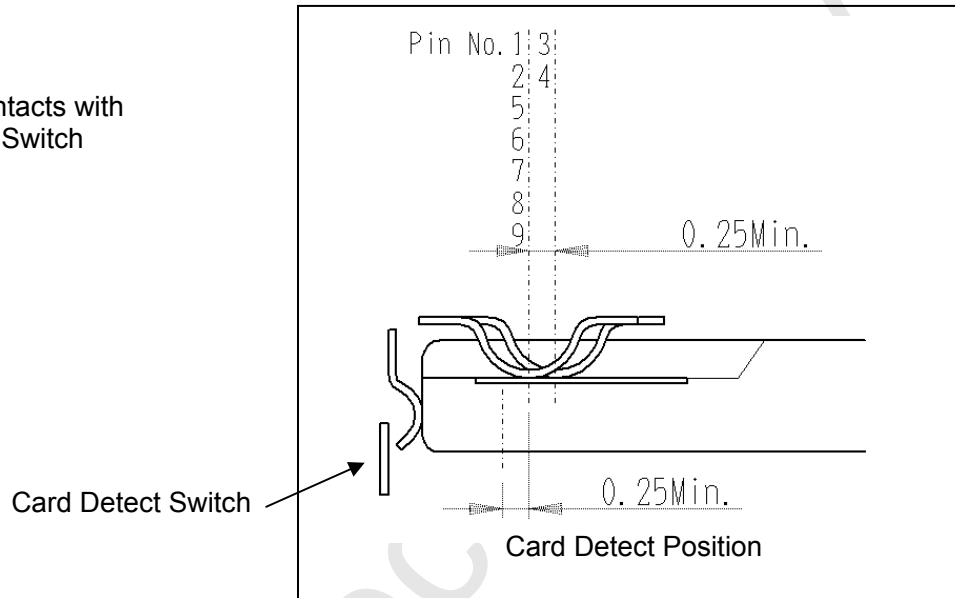
Table 4-2 : Harsh Environment Durability

4.1.6 Connector Pin Spacing

Figure 4-2 shows position of the connector pin. The card dimension allowance and inclination need not considering.

Type A

Single Row Contacts with
Card Detection Switch



"Card Detect Position" means where the card detection switch detects the existence of the card. Refer to Appendix C about implementation of the card detection switch on the connector.

Type B

Single Row Contacts without
Card Detection Switch

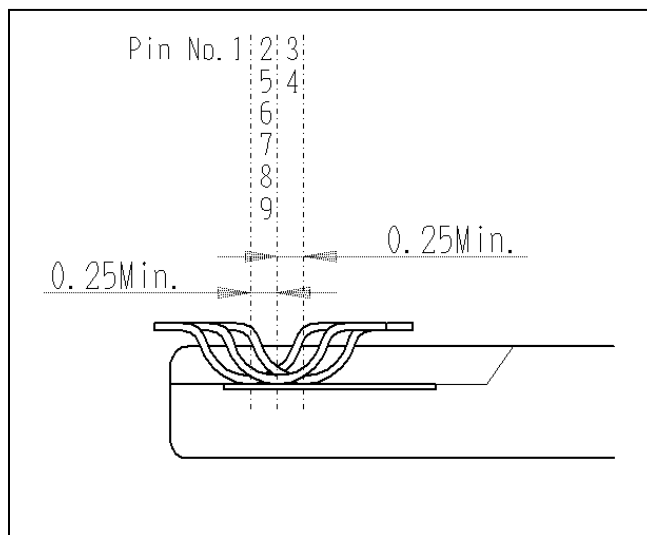


Figure 4-2 : Connector Pin Spacing

4.1.7 Card Over-travel in Push-Push Connector

The push-push connector shall prevent the situation where the contact pins of the connector over-travels the contact pads of the card. During the full insertion cycle the electrical contact to the card shall be maintained.

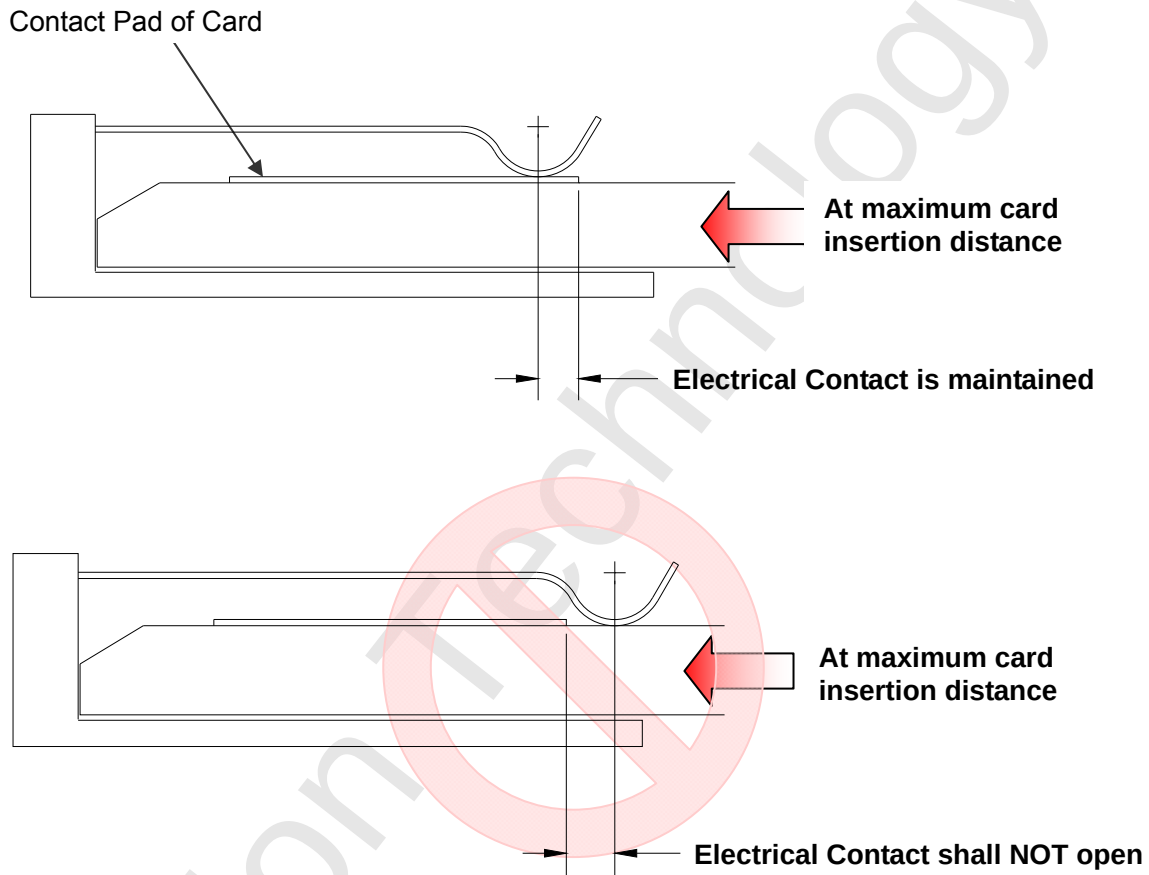


Figure 4-3 : Example of Over-travel Condition

4.2 Card Insertion and Removal

Insertion of the SD Memory Card is possible when the contact area of the card and the contact area of the connector are relatively in the correct position each other. This is ensured by the slanted corners of the card and the connector, respectively.

To guarantee reliable initialization during hot insertion, some measures should be taken on the connector side. It is based on the idea of a defined sequence for card contact connection during the card insertion process. The card contacts are contacted in three steps:

- 1) Ground V_{SS} (pin 3) and supply voltage V_{DD} (pin 4).
- 2) CLK, CMD, DAT0, DAT1, DAT2 and V_{SS} (pin 6).
- 3) CD/DAT3 (pin 1).

Pins 3 and 4 should make contact first when inserting the card and release last when extracting the card.

4.3 Inverse Insertion

Inverse insertion is prevented by the slanted corners of SD Card and connector.

The connector should prevent inverse insertion by detecting the slanted corner of SD Card.

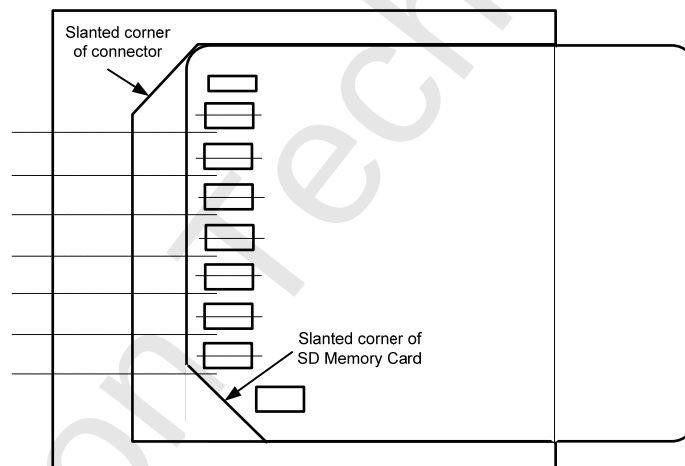


Figure 4-4: Inverse Insertion

5. miniSD to SD Adapter Specification

This chapter describes the detail requirements for Adapters.

Adapter specification is measured with adaptable card inserted.

5.1 Design and Format

Dimensions SD Adapter	24 mm x 32 mm; (min. 23.9 mm x 31.9 mm; max. 24.1 mm x 32.1 mm) Other dimensions Refer to Section 5.3. Testing according to MIL STD 883, Meth 2016
Thickness ¹	'Inter Connect Area': 2.1 mm +/- 0.15 mm Substrate Area': Max 2.25 mm Refer to Figure 5-2 and Figure 5-3.
Label or printable area	In 'Substrate Area' only – Refer to Figure 5-6.
Edges	Smooth edges. Refer to Figure 5-2.
Inverse insertion	Protection on left corner of top view. Refer to Figure 4-4.
Position of ESC contacts	Along middle of shorter edge. Refer to Figure 5-1 and Figure 5-2.

Note 1: Thin card type adapter is not specified because there is no thin card adapter on the market.

Table 5-1 : miniSD to SD Adapter - Dimensions

5.2 Reliability and Durability

Temperature	Operation: -25 deg.C/85 deg.C Storage: -40 deg.C /85 deg.C Storage Test Condition: -40 deg.C (168h)/85 deg.C (500 h)
Moisture and corrosion	Operation: 25 deg.C/95% relative humidity Storage: 40 deg.C/93% relative humidity Storage test condition: 40 deg.C/93% relative humidity/500h Salt water spray: 3% NaCl/35C; 24 h acc. MIL STD Method 1009
Durability	10.000 mating cycles
Bending	10 N (Note 1: Appendix E.1)
Torque	0.15Nm, +/-2.5 deg. max. (Note 1: Appendix E.2)
Drop test	1.5 m free fall
Visual inspection shape and form	No mold skin; complete form; no cavities surface smoothness ≤ -0.1 mm/cm ² within contour; no cracks; no pollution (fat, oil dust, etc.) (Note 1: Appendix E.3 and E.4)
Minimum moving force of WP switch	0.4N min. at any point moving 0mm to 0.4mm from the edge of the card case. (Ensures that the WP switch will not slide while it is inserted connector). (Note 1: Appendix E.5)
WP Switch cycles	Minimum 1000 Cycles (@ Slide force 0.4 N to 5 N)
Rib strength	Tolerance of 4N force 1 second. (Note 2)

Note 1: The test methods are shown in Appendix E.

Note 2: The metal pillar with 1.5mm diameter is stood at right angles to a contact pad. From the side of the middle of each rib, each rib is pushed and applied the force of 4N for 1 second by this pillar. By this pressurization, each rib should not break or junction should not separate. Moreover, after removing pressurization, after 10 minutes each rib should transform and don't overlap the minimum width (1.4mm) contact pad area.

Table 5-2 : Reliability and Durability

This chapter describes the mechanical and electromechanical features of the miniSD Card Adapter. All technical drafts follow DIN ISO standard.



Standard Size SD Card Mechanical Addendum Version 4.00

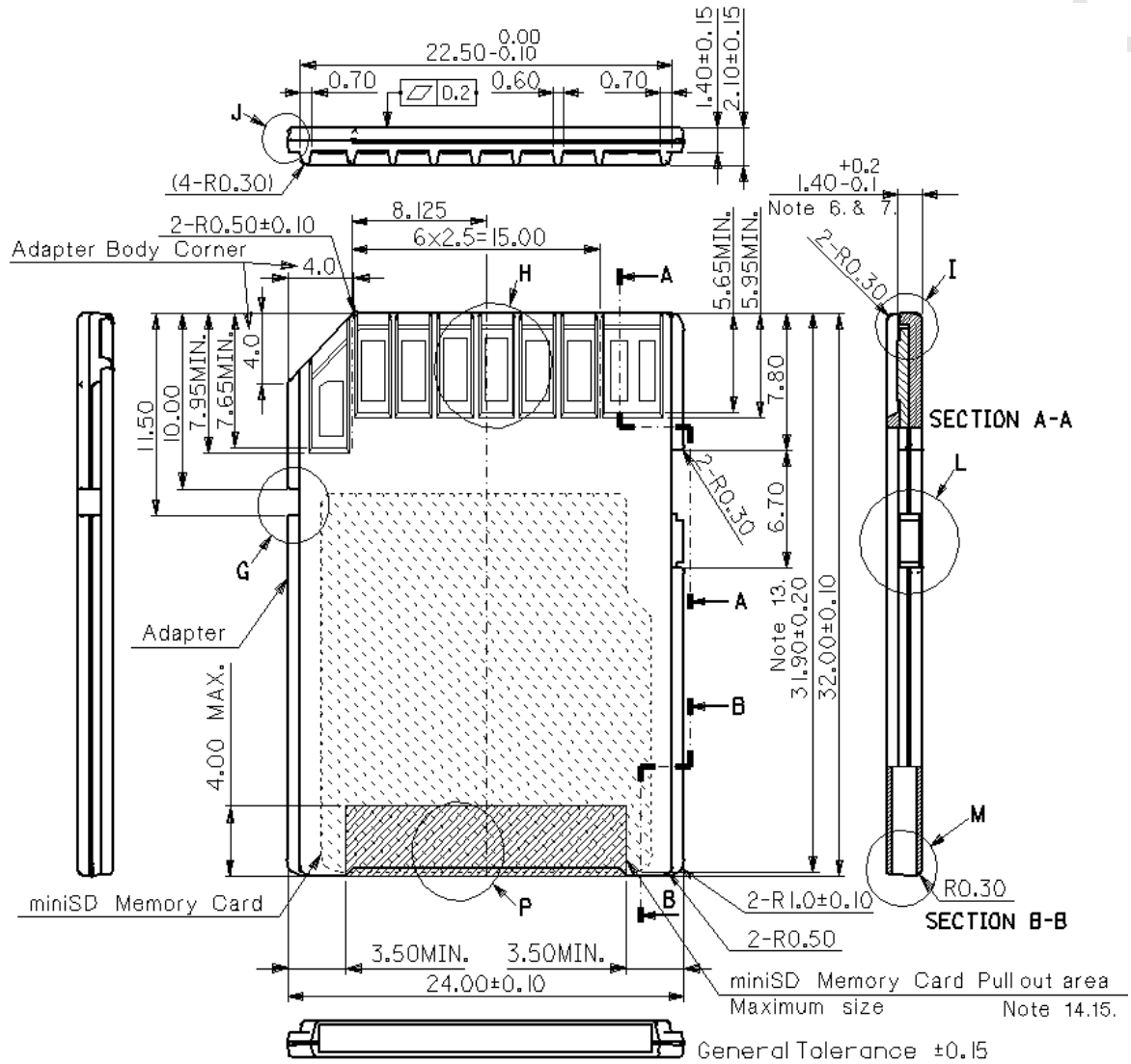
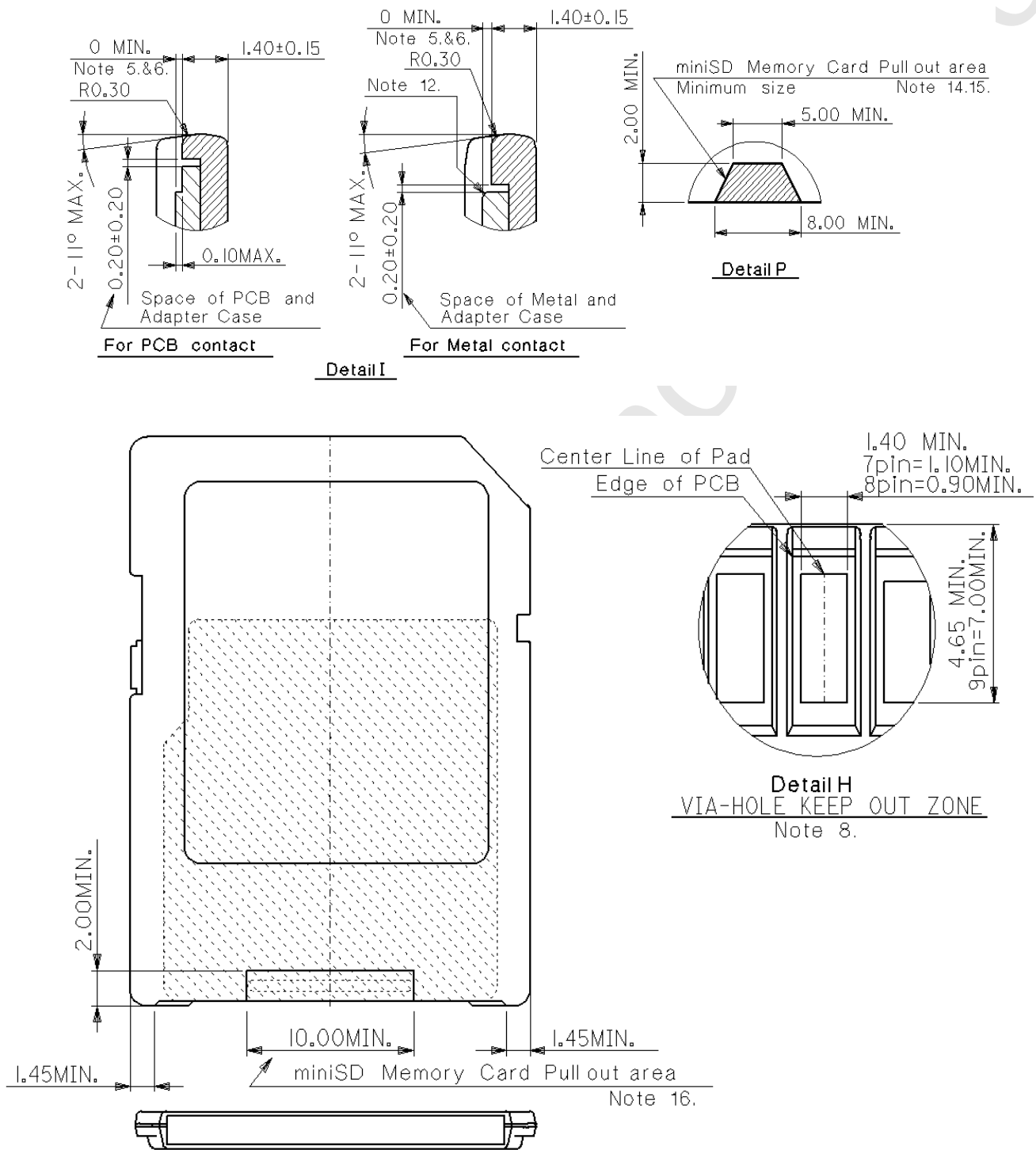
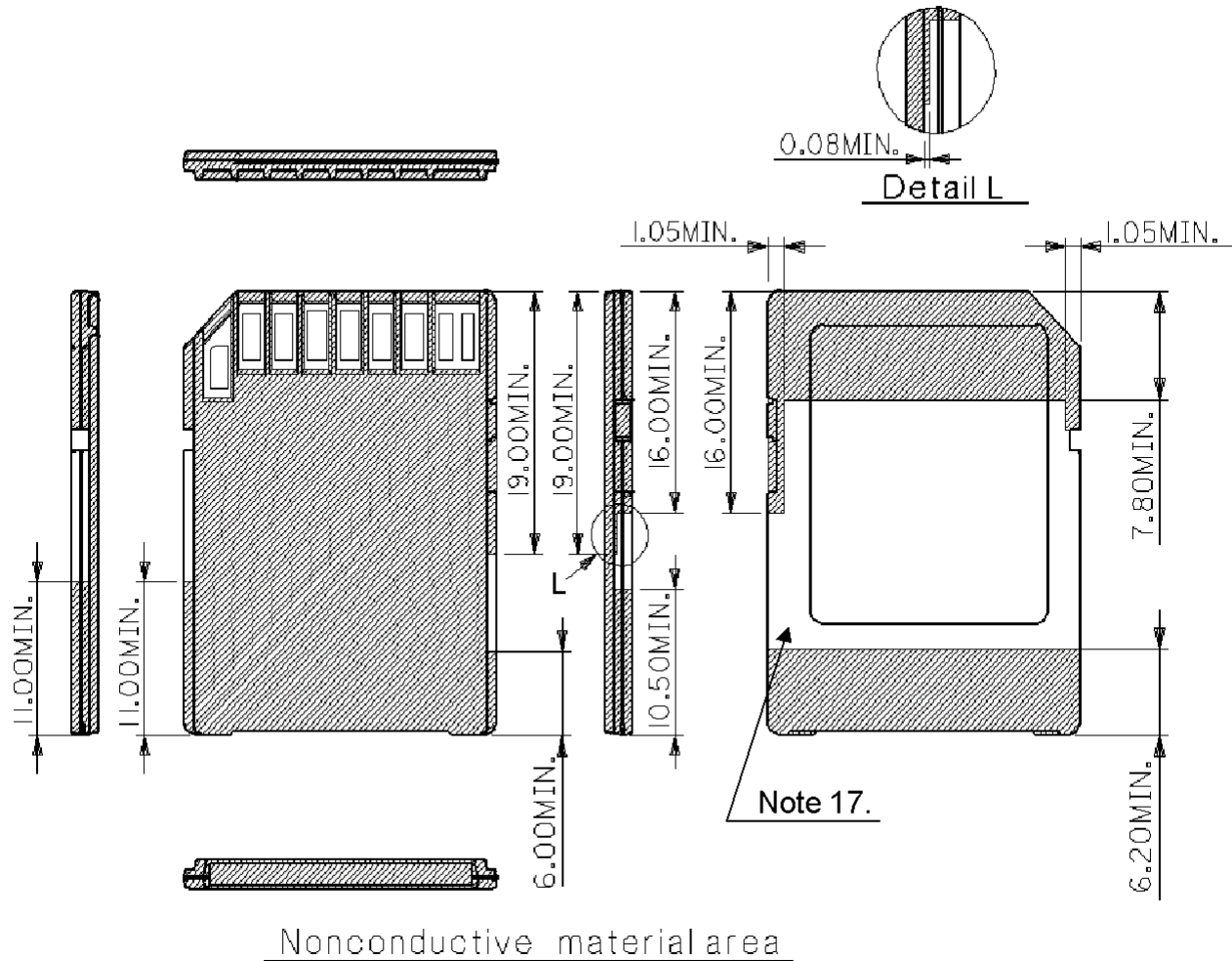


Figure 5-2 : Adapter Dimensions (2 out of 3)

Standard Size SD Card Mechanical Addendum Version 4.00**Figure 5-3 : Adapter Dimensions (3 out of 3)**



Note: Conductive area (except for pads) should be connected to Vss.

Figure 5-4 : Adapter, Nonconductive Areas

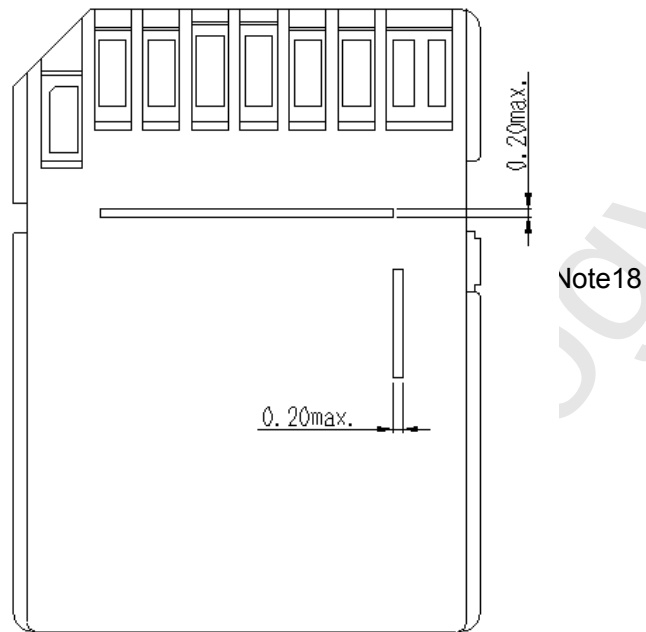
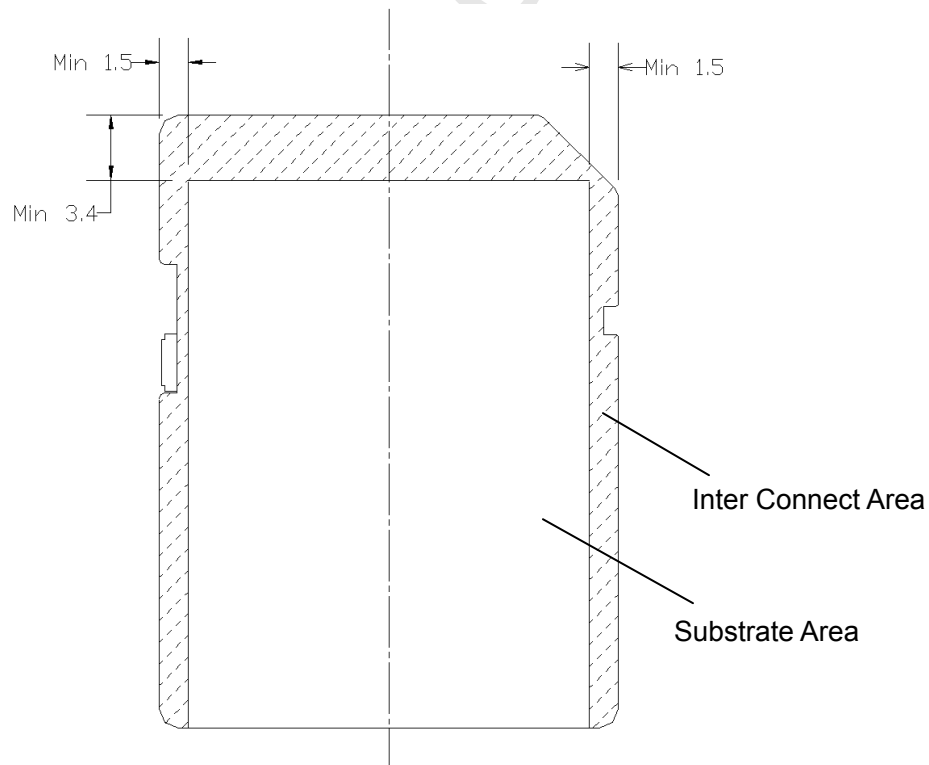


Figure 5-5 : Adapter, Maximum Gap & Hole Dimensions



Note: Refer to the definition of 'substrate' and 'Inter Connect' areas

Figure 5-6 : SD Adapter - Mechanical Description

Notes:

- 1: Refer to Table 3-2 about minimum moving force of WP-key.
- 2: The adapter width of the rest of this hatched area should be 24.0 ± 0.1 .
- 3: Figure 5-1 shows the dimension when the key is pressed towards the direction of each position.
- 4: This detail figure specification shall be applied for the 3-sides of the adapter excluding detail I.
- 5: This dimension is from the surface of the contact pad to the adapter case.
- 6: These dimensions shall be satisfied when no force is put on any pad (0 N) and also when a 0.5N force is applied to the measure point.
- 7: This dimension is from the surface of the contact pad to the surface of the label side of adapter case.
- 8: The via-hole should not be positioned within hatched area.
- 9: The preceding detailed dimensions are to be used for a nonconductive material type WP-key for all types of the adapter. If a conductive material required, detailed dimensions should be submitted for review to the SSTG, by the interested party.
- 10: The write protect key may not protrude from the adapter surface. The key shall retreat inside of the surface with less than 0.3N force, if the key protrudes from the surface.
- 11: WP-key is either a moving type or fixed type. When WP-key is a fixed type, the WP-key shall be in the Write Enable position.
- 12: No burr and no square edge. Recommended to have R0.05mm MIN. or C0.05mm MIN. or suitable similar form.
- 13: This dimension applies when miniSD Card is fully inserted into the adapter case.
- 14: miniSD Card Pull out area (Detail P) should fit within the Maximum Pull out area. The center of Pull out area may be different than adapter center.
- 15: This feature applies only to the bottom surface pull out area (contact pad side).
- 16: This feature applies only to the top surface pull out area (label side).
- 17: For metal cover adapters, the metal cover should be connected directly to Vss.
- 18: If the adapter has gap and hole on bottom side, their width shall be 0.20 mm max. as shown in Figure 5-5.
- 19: The gap dimension of side is indicated in the drawing.

5.4 miniSD to SD Adapter Reliability

Performance and test method described below are characteristic items of miniSD to SD Card Adapter.

Notes

1. The requirements in this section take precedence over requirements referred to in other sections.
2. All testing shall be performed with the miniSD Card inserted.
You may use either a real miniSD Card or/and a dummy test card, if it complies with the miniSD Card specifications. The card may be supplied by a Card vender (preferable) or self built by a connector manufacturer.

Temperature	15deg.C to 35deg.C
Air pressure	86 to 106 kPa
Relative humidity	25% to 85%

If conditions shall be closely controlled in order to obtain reproducible results, the parameters shall be:

Temperature	23deg.C +/- 1deg.C
Air pressure	86 to 106 kPa
Relative humidity	50% +/- 2%

durability	Adapter & Host Connector:10.000 mating cycles Adapter & miniSD Card:10.000 mating cycles
Minimum moving force of WP switch	0.4N Minimum at 0.4mm

5.4.1 miniSD to SD Adapter Mechanical Performance

This section specifies the mechanical performance of the Standard Size SD Card Adapter.

Office Environment

STANDARD	TESTING
Guaranteed number of insertions/extractions =10,000 minimum	Refer to Office Environment, Section 5.4.5

Harsh Environment

STANDARD	TESTING
Guaranteed number of insertions/extractions = 5,000 minimum	Refer to Harsh Environment, Section 5.4.5

miniSD Card Pulling Force

STANDARD	TESTING
The card extraction force shall be 1.5 N minimum.	Pull force to be measured after 10,000 mating cycles. Extraction speed of 25mm/minute. (400-600 cycles/hour).

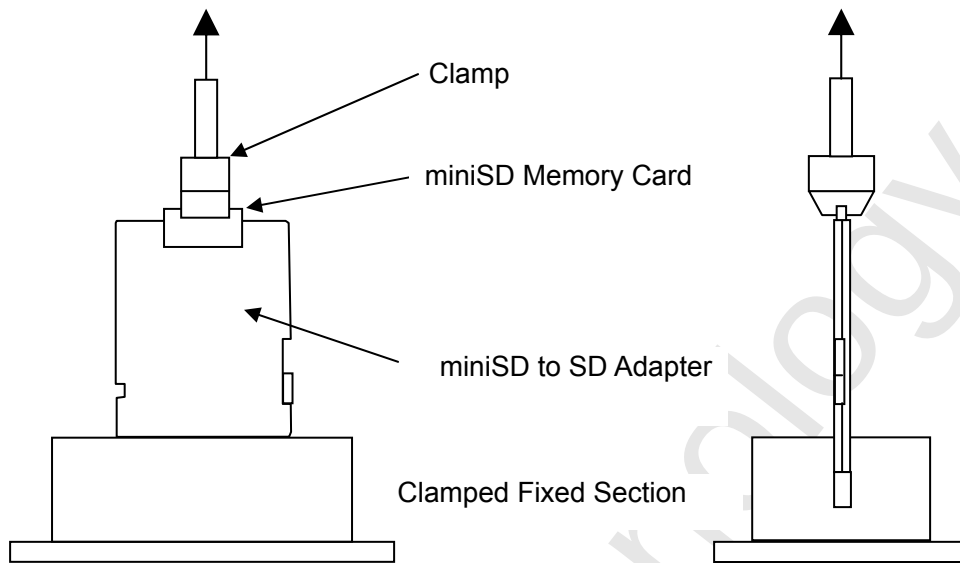


Figure 5-7 : Card Pulling Force Test Method

Voltage Proof

STANDARD	TESTING
a. No shorting or other damages when 500 Vrms AC is applied for 1 minute	IEC 60512-4-1
b. Current leakage 1 mA maximum	

Insulation Resistance

STANDARD	TESTING
a. Initially 1000 M-ohm minimum	IEC 60512-3-1. Measure within 1 minute after applying 500V DC.
b. After test 100 M-ohm minimum	

Temperature Rise

STANDARD	TESTING
0.5 A per contact	IEC 60512-5-1. Based upon 30deg.C rise above ambient temperature.

5.4.2 miniSD to SD Adapter Environmental Performance**Operating Environment**

STANDARD
Operating Temperature: -25deg.C to +85deg.C
Relative humidity: 95% maximum (non-condensing)

Storage Environment

STANDARD
Storage Temperature: -40deg.C to +85deg.C
Relative humidity: 95% maximum (non-condensing)

5.4.3 miniSD to SD Adapter Environmental Resistance

Damp Heat, Cyclic

STANDARD	TESTING
Per Contact Resistance (low level) Section, Part b Per Insulation Resistance Section, Part b	IEC 60512-11-12 10 cycles (1 cycle = 24 hours) with connectors engaged

Rapid Change Of Temperature

STANDARD	TESTING
No physical damage shall occur during testing Per Contact Resistance (low level) Section, Part b Per Insulation Resistance Section, Part b	IEC 60512-11-4 -55deg.C to +85deg.C 5 minute transition time (max) 5 cycles (1 cycle = 1 hour) with connectors engaged

Dry Heat

STANDARD	TESTING
Per Contact Resistance (low level) Section, Part b	IEC 60512-11-9 85deg.C, 96 hours with connectors engaged. Exclude load and insulation resistance measurements

Vibration

STANDARD	TESTING
a. No mechanical damage shall occur on the parts b. Shall not cause current interruption greater than 100 ns c. miniSD Card shall not pull away from the Adapter	IEC 60512-6-4 20 m/s ² (2G) peak amplitude, 10 Hz to 2000 Hz, 5 minutes per 1 cycle, 10 cycle per 1 axis total 30 cycles per 3 axis.

Note: Method of measuring discontinuity associated with connector: At DC 5V and 150mA Max. ; decent of voltage more than 50%(=less than 2.5V) as discontinuity. Discontinuity specification for the connector: Maximum 100nsec. of discontinuity period.

Shock

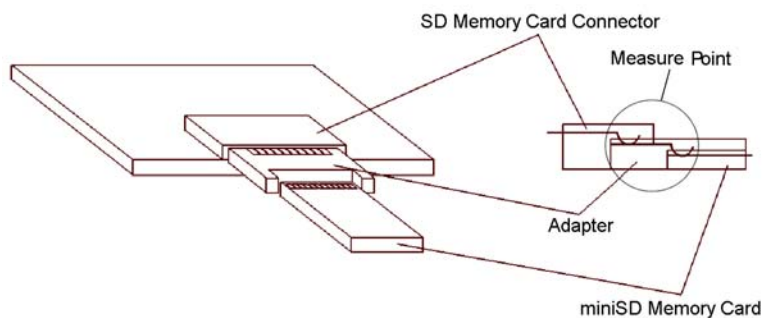
STANDARD	TESTING
a. No mechanical damage shall occur on the parts b. Shall not cause current interruption greater than 100 ns c. miniSD Card shall not pull away from the Adapter	IEC 60512-6-3 Acceleration 490 m/s ² (50G) Standard holding time 11 ms, Semi-sine wave Velocity change 3.44 m/s (11.3 ft/s)

Note: Method of measuring discontinuity associated with connector: At DC 5V and 150mA Max. ; decent of voltage more than 50%(=less than 2.5V) as discontinuity. Discontinuity specification for the connector: Maximum 100nsec. of discontinuity period.

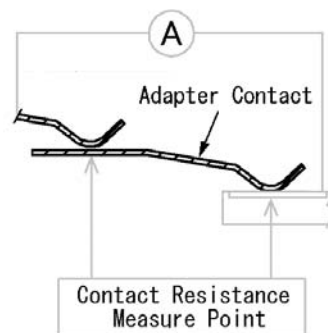
5.4.4 miniSD to SD Adapter Electrical Performance

Contact Resistance-Millivolt Level Method

STANDARD	TESTING
a. Initially 100 milli-ohm maximum	IEC 60512-2-1 Open voltage 20 mV Test current 1 mA
b. After test 40 milli-ohm maximum change	a. Measure and record the initial resistance (R_i) of the separate connector contact interface. Refer to Figure 5-8. R_i 100 milli-ohm b. Measure and record the resistance after test (R_f) of the connector system. Resistance value after test: $R_f = R_i \pm 40$ milli-ohm



Example of Measurement Fixture



Detail of Measure Point

Figure 5-8 : Contact Resistance Measurement method

Cold

STANDARD	TESTING
Per Contact Resistance (low level) Section, Part b	IEC 60512-11-10 -25deg.C, 96 hours with connectors engaged

Damp Heat, Steady State

STANDARD	TESTING
Per Contact Resistance (low level) Section, Part b	IEC 60512-11-3 Steady State 40deg.C, 90 to 95% RH
Per Insulation Resistance Section, Part b	96 hours with connectors engaged

Hydrogen Sulfide

STANDARD	TESTING
Per Contact Resistance (low level) Section, Part b	JEIDA 38 3 PPM hydrogen sulfide 40deg.C, approx. 80% RH 96 hours, with connectors engaged

5.4.5 miniSD to SD Adapter Environmental Durability

Test conditions for the mate/unmate cycles are:

Cycle Rate 400-600 cycles per hour
 Temperature 15deg.C to 35deg.C
 Relative Humidity 25% to 85%
 Air Pressure 86 to 106 kPa

Office Environment

The office environment is defined in EIA-364-B Class 1.1 - year round air conditioning (non-filtered) with humidity control.

Contact resistance - Part a
Mate and unmate the connector for a total of 10,000 cycles ^{Note}
Contact resistance - Part b

Note: After each 10 cycles stop the insertion and rest the connector for 5 to 10 minutes. Total 10,000 cycles. Mate and unmate speed is less than 10cycles per 1 minute.

Harsh Environment

The harsh environment is defined in EIA-364-B Class 1.3—no air conditioning, no humidity control with normal heating and ventilation:

Contact resistance - Part a	
Mate and unmate the connector 1,000 cycles ^{Note}	TOTAL CYCLES = 1,000
Damp heat, cyclic Section (1 cycle=24hours)	Total 1 cycle
Mate and unmate the connector 1,000 cycles ^{Note}	TOTAL CYCLES = 2,000
Damp heat, cyclic Section (1 cycle=24hours)	Total 2 cycle
Mate and unmate the connector 3,000 cycles ^{Note}	TOTAL CYCLES = 5,000
Damp heat, cyclic Section (1 cycle=24hours)	Total 3 cycle
Hydrogen sulfide per Hydrogen Sulfide Section	96 hours
Contact resistance - Part b	

Note: After each 10 cycles stop the insertion and rest the connector for 5 to 10 minutes. Total 10,000 cycles. Mate and unmate speed is less than 10cycles per 1 minute.

6. microSD to SD Adapter Specification

6.1 Design and Format

Dimensions SD Adapter	24 mm x 32 mm; (min. 23.9 mm x 31.9 mm; max.24.1 mm x 32.1 mm) Other dimensions: Refer to Figure 6-1 through Figure 6-3. Testing according to MIL STD 883, Meth 2016
Thickness	'Inter Connect Area': 2.1 mm +/- 0.15 mm 'Substrate Area': Max 2.25 mm Refer to Figure 6-2 and Figure 6-3.
Label or printable area	In 'Substrate Area' only – Refer to Figure 5-6.
Edges	Smooth edges Refer to Figure 6-2.
Inverse insertion	Protection on left corner (top view). Refer to Figure 4-4.
Position of ESC contacts	Along middle of shorter edge. Refer to Figure 6-1 and Figure 6-2.

Table 6-1 : microSD to SD Adapter - Dimensions

6.2 Reliability and Durability

Temperature	Operation Temperature: -25 to 85 deg.C Storage Temperature : -40 to 85 deg.C Storage Temperature test condition: -40 deg.C (168h)/85 deg.C (500 h)
Moisture and corrosion	Operation: 25 deg.C /95% relative humidity Storage: 40 deg.C /93% relative humidity Storage Temperature test condition: 40 deg.C /93% relative humidity/500h Salt water spray: 3% NaCl/35C; 24 h acc. MIL STD Method 1009
Durability	10.000 mating cycles
Bending	10 N (Note 1: Appendix E.1)
Torque	0.15Nm, +/-2.5 deg. max. (Note 1: Appendix E.2)
Drop test	1.5 m free fall
Visual inspection shape and form	No mold skin; complete form; no cavities surface smoothness ≤ -0.1 mm/cm ² within contour; no cracks; no pollution (fat, oil dust, etc.) (Note 1: Appendix E.3 and E.4)
Minimum moving force of WP switch	0.4N min. at any point moving 0mm to 0.4mm from the edge of the card case. (Ensures that the WP switch will not slide while it is inserted to the connector). (Note 1: Appendix E.5)
WP Switch cycles	Minimum 1000 Cycles (@ Slide force 0.4 N to 5 N)
Rib strength	Tolerance of 4N force 1 second. (Note 2)

Note 1: The test methods are shown in Appendix E.

Note 2: The metal pillar with 1.5mm diameter is stood at right angles to a contact pad. From the side of the middle of each rib, each rib is pushed and applied the force of 4N for 1 second by this pillar. By this pressurization, each rib should not break or junction should not separate. Moreover, after removing pressurization, after 10 minutes each rib should transform and don't overlap the minimum width (1.4mm) contact pad area.

Table 6-2 : Reliability and Durability

6.3 microSD to SD Card Adapter Mechanical Dimensions

This Section describes the mechanical and electromechanical features of the microSD Card Adapter. All technical drafts follow DIN ISO standard.

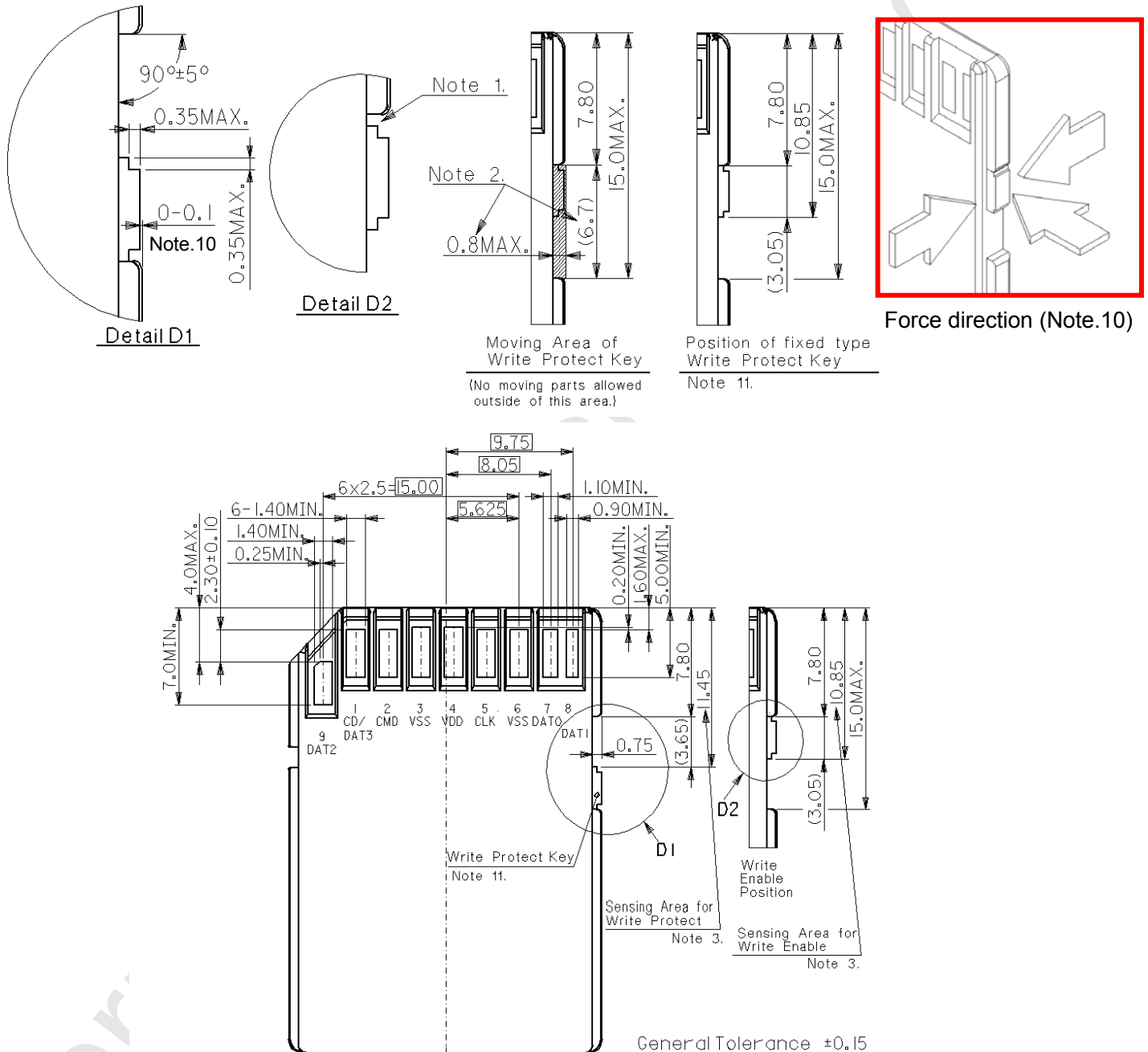


Figure 6-1 : SD Adapter Dimensions (1 out of 3)

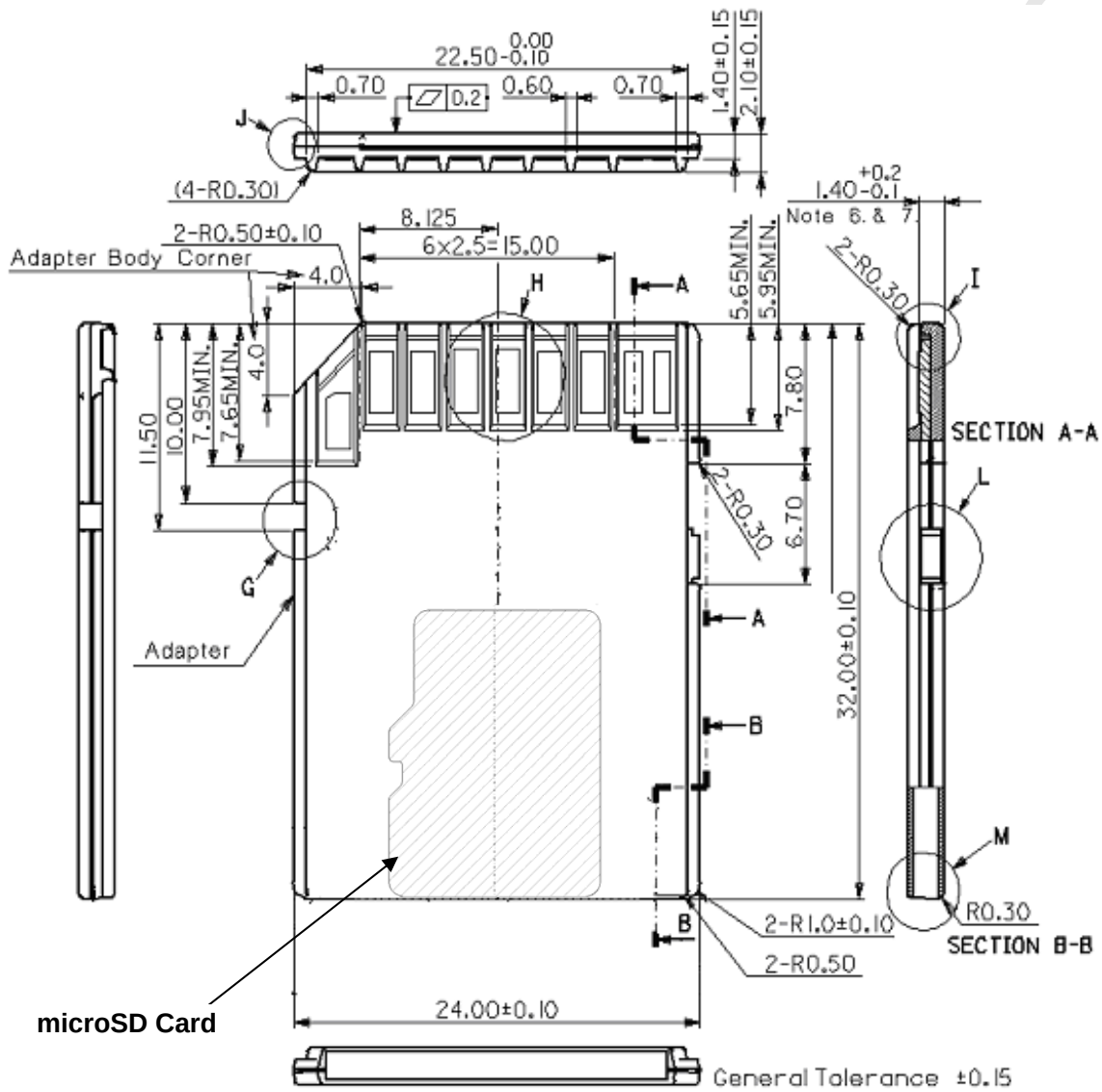
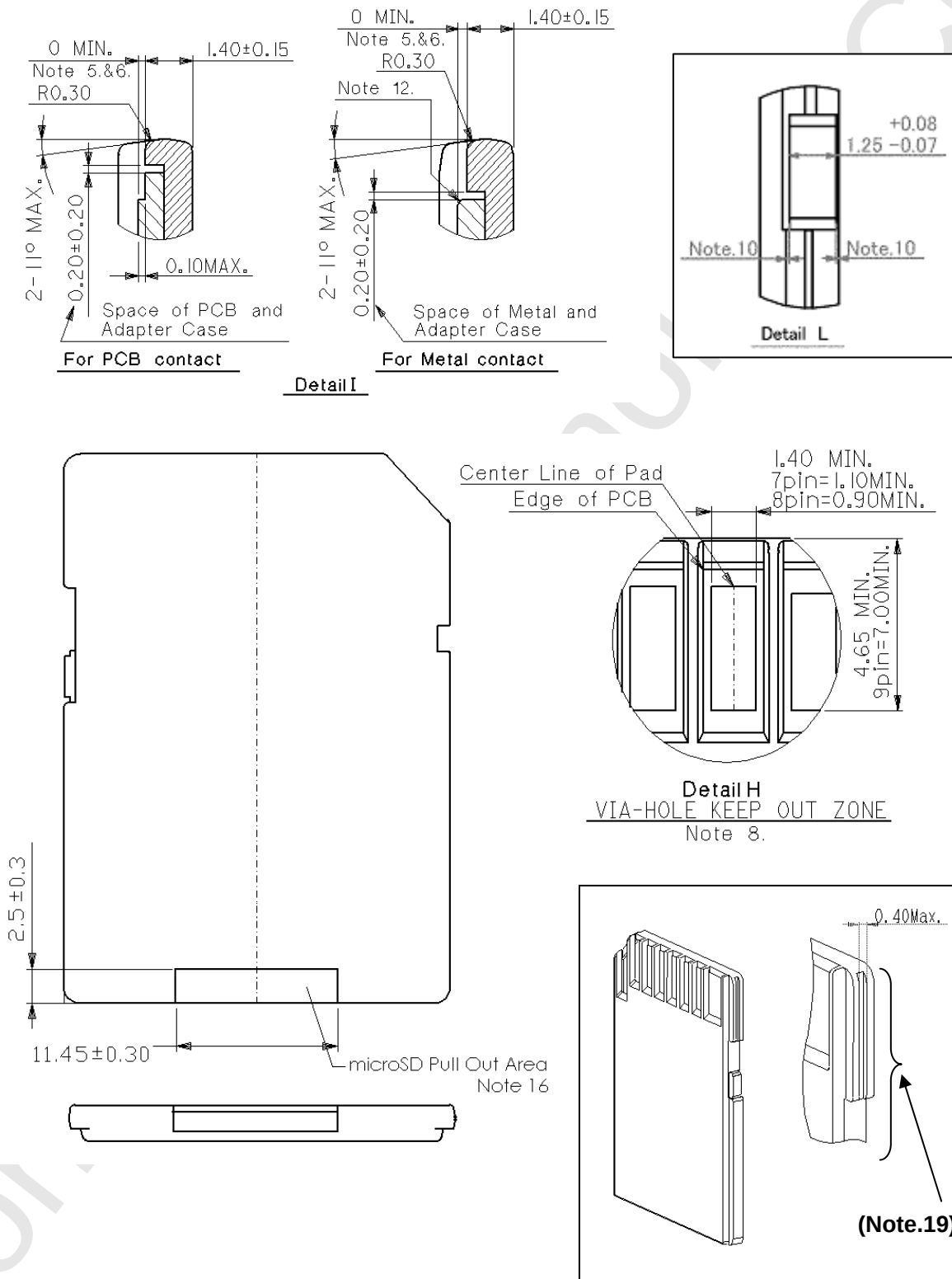
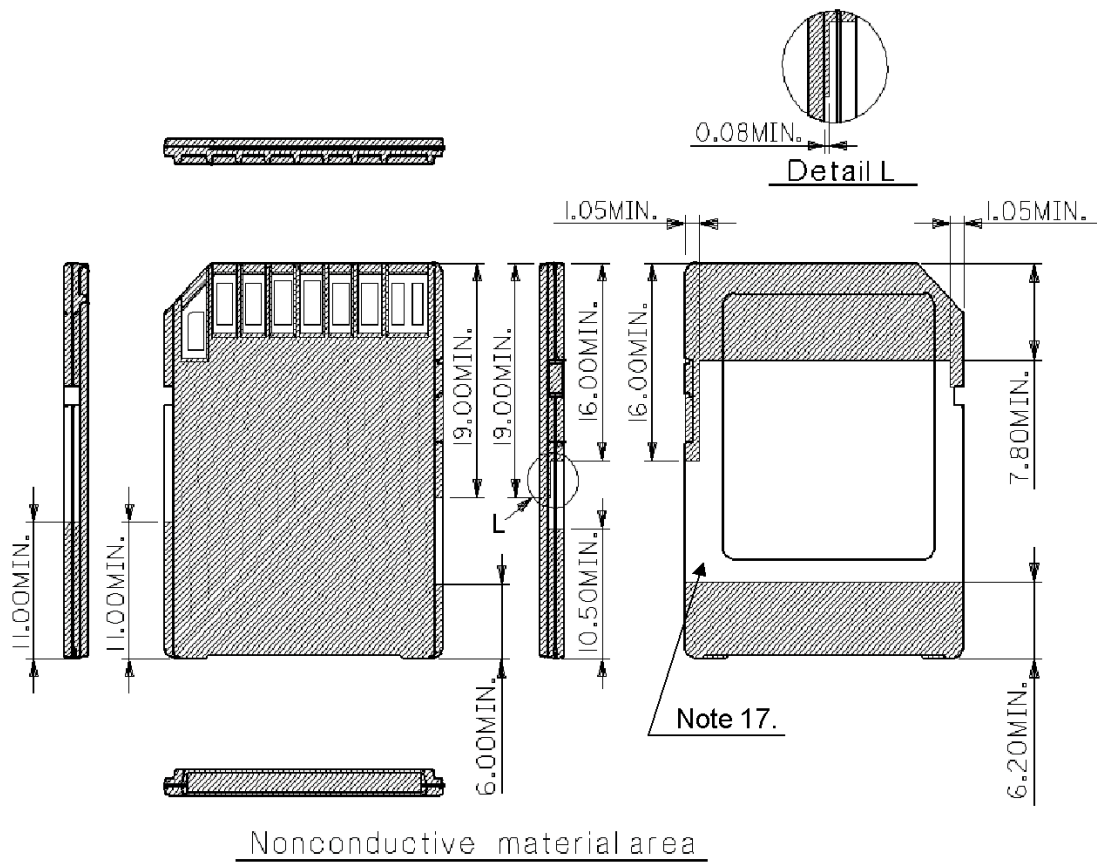


Figure 6-2 : SD Adapter Dimensions (2 out of 3)

Standard Size SD Card Mechanical Addendum Version 4.00**Figure 6-3 : SD Adapter Dimensions (3 out of 3)**



Note: Conductive area (except for pads) should be connected to Vss.

Figure 6-4 : SD Adapter Non-Conductive Areas

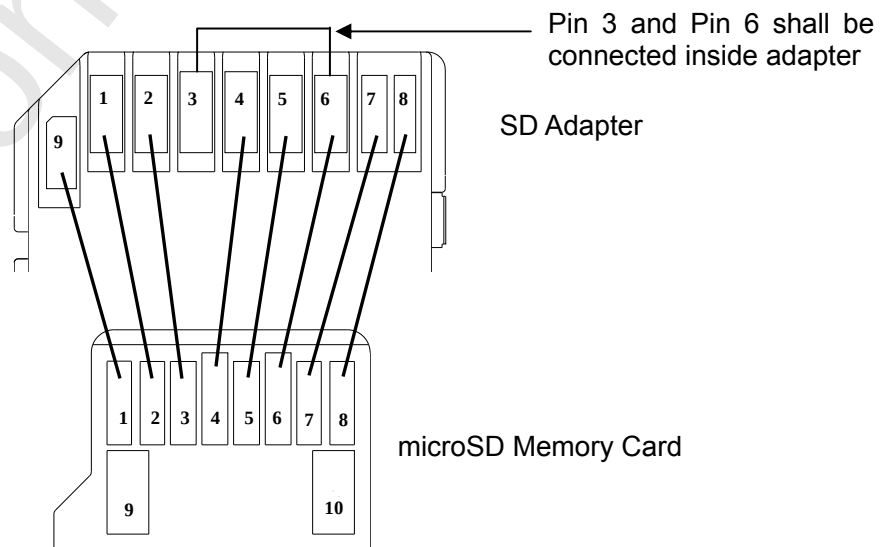


Figure 6-5 : SD Adapter Interconnect

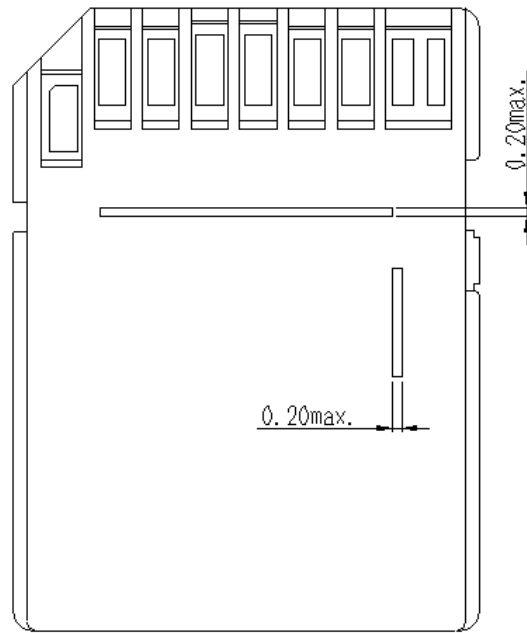


Figure 6-6 : Adapter, Maximum Gap & Hole Dimensions

Notes:

- 1: Refer to Table 3-2 about minimum moving force of WP-key.
- 2: The adapter width of the rest of this hatched area should be 24.0 ± 0.1 mm.
- 3: The figure shown on the drawing is the dimension when the key is pressed towards the direction of each position.
- 4: This detail figure specification shall be applied for the 3-sides of the adapter excluding detail I.
- 5: This dimension is from the surface of the contact pad to the adapter case.
- 6: These dimensions shall be satisfied when no force is put on any pad (0 N) and also when a 0.5N force is applied to the measure point.
- 7: This dimension is from the surface of the contact pad to the surface of the label side of adapter case.
- 8: The via-hole should not be positioned within hatched area.
- 9: The preceding detailed dimensions are to be used for a nonconductive material type WP-key for all types of the adapter. If a conductive material required, detailed dimensions should be submitted for review to the SSTG, by the interested party.
- 10: The write protect key may not protrude from the adapter surface. The key shall retreat inside of the surface with less than 0.3N force, if the key protrudes from the surface.
- 11: WP-key is either a moving type or fixed type. When WP-key is a fixed type, the WP-key shall be in the Write Enable position.
- 12: No burr and no square edge. Recommended to have R0.05mm MIN. or C0.05mm MIN. or suitable similar form.
- 13: This dimension applies when miniSD Card is fully inserted into the adapter case.
- 14: microSD Card Pull out area (Detail P) should fit within the Maximum Pull out area. The center of Pull out area may be different than adapter center.
- 15: This feature applies only to the bottom surface pull out area (contact pad side).
- 16: This feature applies only to the top surface pull out area (label side).
- 17: For metal cover adapters, the metal cover should be connected directly to Vss.
- 18: If the adapter has gap and hole on bottom side, their width shall be 0.20 mm max. as shown in Figure 6-6.
- 19: The gap dimension of side is indicated in the drawing.

6.4 microSD to SD Adapter Reliability

Performance and test method described below are characteristic items of microSD to SD Card Adapter.

Notes

1. The requirements in this section take precedence over requirements referred to in other sections.
2. All testing shall be performed with the microSD Card inserted.
You may use either a real microSD Card or/and a dummy test card, if it complies with the microSD Card specifications. The card may be supplied by a Card vender (preferable) or self built by a connector manufacturer.

Temperature	15deg.C to 35deg.C
Air pressure	86 to 106 kPa
Relative humidity	25% to 85%

If above conditions are too moderate to obtain reproducible results, the following conditions should be used.

Temperature	23deg.C +/- 1deg.C
Air pressure	86 to 106 kPa
Relative humidity	50% +/- 2%

durability	Adapter & Host Connector:10.000 mating cycles Adapter & microSD Card:10.000 mating cycles
Minimum moving force of WP switch	0.4N Minimum at 0.4mm

6.4.1 microSD to SD Adapter Mechanical Performance

This section specifies the mechanical performance of the Standard Size SD Card Adapter.

Office Environment

STANDARD	TESTING
Guaranteed number of insertions/extractions =10,000 minimum	Refer to Office Environment in Section 6.4.5.

Harsh Environment

STANDARD	TESTING
Guaranteed number of insertions/extractions = 3,000 minimum	Refer to Harsh Environment in Section 6.4.5.

microSD Card Pulling Force

STANDARD	TESTING
The card extraction force shall be: 1.09N minimum.	Pull force to be measured after 10,000 mating cycles. Extraction speed of 25mm/minute. (400-600 cycles/hour).

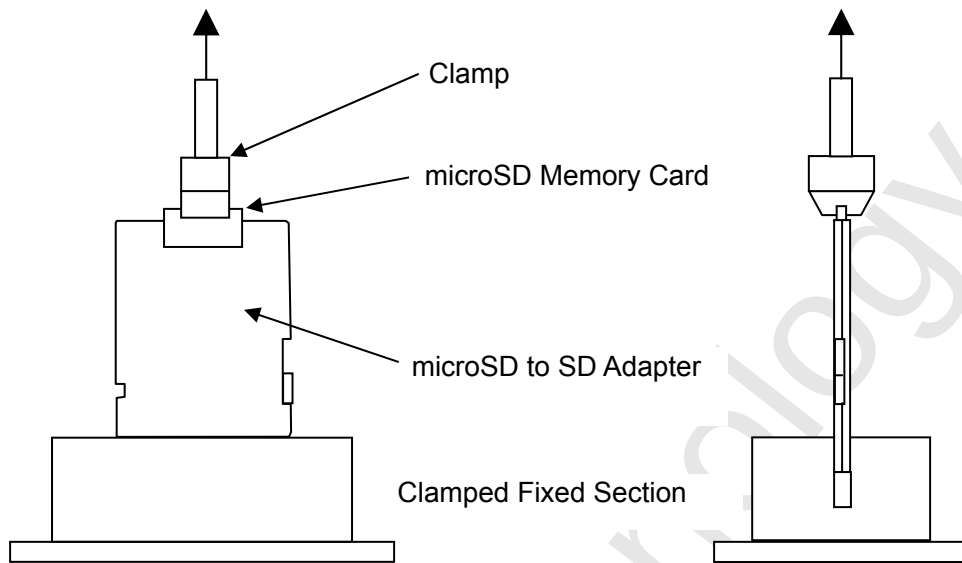


Figure 6-7 : Card Pulling Force Test Method

Voltage Proof

STANDARD	TESTING
a. No shorting or other damages when 500 Vrms AC is applied for 1 minute	IEC 60512-4-1
b. Current leakage 1 mA maximum	

Insulation Resistance

STANDARD	TESTING
a. Initially 1000 M-ohm minimum	IEC 60512-3-1. Measure within 1 minute after applying 500V DC.
b. After test 100 M-ohm minimum	

Temperature Rise

STANDARD	TESTING
0.5 A per contact	IEC 60512-5-1. Based upon 30deg.C rise above ambient temperature.

6.4.2 microSD to SD Adapter Environmental Performance**Operating Environment**

STANDARD
Operating Temperature: -25deg.C to +85deg.C
Relative humidity: 95% maximum (non-condensing)

Storage Environment

STANDARD
Storage Temperature: -40deg.C to +85deg.C
Relative humidity: 95% maximum (non-condensing)

6.4.3 microSD to SD Adapter Environmental Resistance

Damp Heat, Cyclic

STANDARD	TESTING
Per Contact Resistance (low level) Section, Part b Per Insulation Resistance Section, Part b	IEC 60512-11-12 10 cycles (1 cycle = 24 hours) with connectors engaged

Rapid Change Of Temperature

STANDARD	TESTING
No physical damage shall occur during testing Per Contact Resistance (low level) Section, Part b Per Insulation Resistance Section, Part b	IEC 60512-11-4 -55deg.C to +85deg.C 5 minute transition time (max) 5 cycles (1 cycle = 1 hour) with connectors engaged

Dry Heat

STANDARD	TESTING
Per Contact Resistance (low level) Section, Part b	IEC 60512-11-9 85deg.C, 96 hours with connectors engaged. Exclude load and insulation resistance measurements

Vibration

STANDARD	TESTING
a. No mechanical damage shall occur on the parts b. Shall not cause current interruption greater than 100 ns c. microSD Card shall not pull away from the Adapter	IEC 60512-6-4 20 m/s ² (2G) peak amplitude, 10 Hz to 2000 Hz, 5 minutes per 1 cycle, 10 cycle per 1 axis total 30 cycles per 3 axis.

Note: Method of measuring discontinuity associated with connector: At DC 5V and 150mA Max. ; decent of voltage more than 50%(=less than 2.5V) as discontinuity. Discontinuity specification for the connector: Maximum 100nsec. of discontinuity period.

Shock

STANDARD	TESTING
a. No mechanical damage shall occur on the parts b. Shall not cause current interruption greater than 100 ns c. microSD Card shall not pull away from the Adapter	IEC 60512-6-3 Acceleration 490 m/s ² (50G) Standard holding time 11 ms, Semi-sine wave Velocity change 3.44 m/s (11.3 ft/s)

Note: Method of measuring discontinuity associated with connector: At DC 5V and 150mA Max. ; decent of voltage more than 50%(=less than 2.5V) as discontinuity. Discontinuity specification for the connector: Maximum 100nsec. of discontinuity period.

6.4.4 microSD to SD Adapter Electrical Performance

Contact Resistance-Millivolt Level Method

STANDARD	TESTING
a. Initially 100 milli-ohm maximum	IEC 60512-2-1 Open voltage 20 mV Test current 1 mA
b. After test 40 milli-ohm maximum change	a. Measure and record the initial resistance (R_i) of the separate connector contact interface. Refer to Figure 6-8. R_i 100 milli-ohm b. Measure and record the resistance after test (R_f) of the connector system. Resistance value after test: $R_f = R_i \pm 40$ milli-ohm

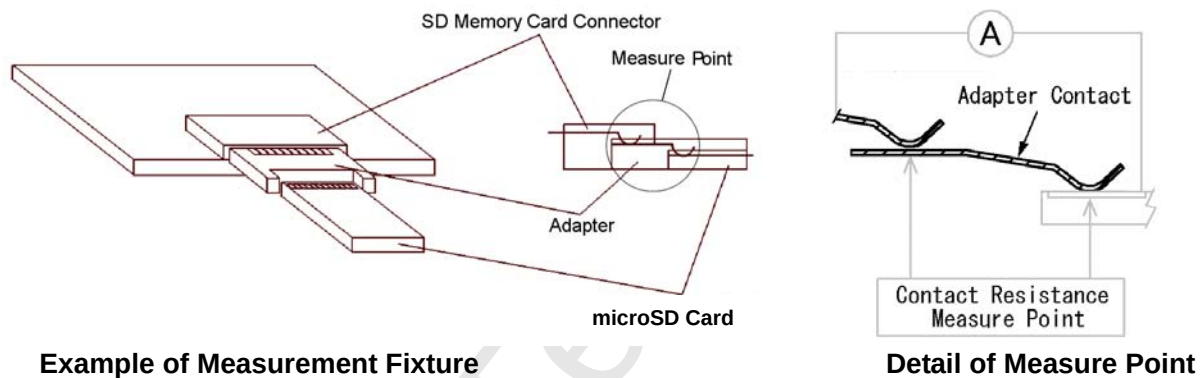


Figure 6-8 : Contact Resistance Measurement method

Cold

STANDARD	TESTING
Per Contact Resistance (low level) Section, Part b	IEC 60512-11-10 -25deg.C, 96 hours with connectors engaged

Damp Heat, Steady State

STANDARD	TESTING
Per Contact Resistance (low level) Section, Part b	IEC 60512-11-3 Steady State 40deg.C, 90 to 95% RH
Per Insulation Resistance Section, Part b	96 hours with connectors engaged

Hydrogen Sulfide

STANDARD	TESTING
Per Contact Resistance (low level) Section, Part b	JEIDA 38 3 PPM hydrogen sulfide 40deg.C, approx. 80% RH 96 hours, with connectors engaged

6.4.5 microSD to SD Adapter Environmental Durability

Test conditions for the mate/unmate cycles are:

Cycle Rate 400-600 cycles per hour
 Temperature 15deg.C to 35deg.C
 Relative Humidity 25% to 85%
 Air Pressure 86 to 106 kPa

Office Environment

The office environment is defined in EIA-364-B Class 1.1 - year round air conditioning (non-filtered) with humidity control.

Contact resistance - Part a
Mate and unmate the connector for a total of 10,000 cycles ^{Note}
Contact resistance - Part b

Note: After each 10 cycles stop the insertion and rest the connector for 5 to 10 minutes.

Air blow card (dry air) for 3secs:

at each 100 cycle interval (10 times) from start to 1000 cycles.

at each 1000 cycle interval (9 times) from 1001 to 10,000 cycles.

Mate and unmate speed is less than 10cycles per 1 minute.

Harsh Environment

The harsh environment is defined in EIA-364-B Class 1.3—no air conditioning, no humidity control with normal heating and ventilation:

Contact resistance - Part a	
Mate and unmate the connector 500 cycles ^{Note}	TOTAL CYCLES = 500
Damp heat, cyclic Section (1 cycle=24hours)	Total 1 damp heat cycle
Mate and unmate the connector 500 cycles ^{Note}	TOTAL CYCLES = 1,000
Damp heat, cyclic Section (1 cycle=24hours)	Total 2 damp heat cycles
Mate and unmate the connector 2,000 cycles ^{Note}	TOTAL CYCLES = 3,000
Damp heat, cyclic Section (1 cycle=24hours)	Total 3 damp heat cycles
Hydrogen sulfide per Hydrogen Sulfide Section	96 hours
Contact resistance - Part b	

Note: After each 10 cycles stop the insertion and rest the connector for 5 to 10 minutes.

Air blow card (dry air) for 3secs:

at each 100 cycle interval (10 times) from start to 1000 cycles.

at each 1000 cycle interval (2 times) from 1001 to 3,000 cycles.

Mate and unmate speed is less than 10cycles per 1 minute.

Appendix A (Normative) : Reference

A.1 Reference

This specification refers the following documents.

- | | | |
|------|--|---|
| (1) | SD Specifications Part 1 Physical Layer Specifications Version 3.01 and Version 4.00 | |
| (2) | SD Specifications Part 1 UHS-II Addendum Version 1.00 | |
| (3) | SD/SDA Logo Guideline Version 2.00 May 9, 2006 | |
| (4) | ASME Y14.5M-1994 | Geometric dimensioning and tolerancing |
| (5) | EIA-364-B | EIA Standard Acceleration Test Procedure for Electrical Connectors |
| (6) | ISO/IEC10373-6 | Electrical characteristics of contactless smart cards |
| (7) | ISO 7816-1 | UV light exposure |
| (8) | IEC 60512 (all parts) | Connectors for electronic equipment – Tests and measurements |
| (9) | IEC 60512-2-1 Ed. 1.0:2002 (IEC 512-2-2a) | Electrical continuity and contact resistance tests - Test 2a: Contact resistance - Millivolt level method |
| (10) | IEC 60512-3-1 Ed. 1.0:2002 (IEC 512-2-3a) | Insulation tests - Test 3a: Insulation resistance |
| (11) | IEC 60512-4-1 Ed. 1.0:2003 (IEC 512-2-4a) | Voltage stress tests - Test 4a: Voltage proof |
| (12) | IEC 60512-5-1 Ed. 1.0:2002 (IEC-512-3-5a) | Current-carrying capacity tests - Test 5a: Temperature rise |
| (13) | IEC 60512-6-3 Ed. 1.0:2002 (IEC 512-4-6c) | Dynamic stress tests - Test 6c: Shock |
| (14) | IEC 60512-6-4 Ed. 1.0:2002 (IEC 512-4-6d) | Dynamic stress tests - Test 6d: Vibration (sinusoidal) |
| (15) | IEC 60512-11-3 Ed. 1.0:2002 (IEC 512-6-11c) | Climatic tests - Test 11c: Damp heat, steady state |
| (16) | IEC 60512-11-4 Ed. 1.0:2002 (IEC 512- 6-11d) | Climatic tests - Test 11d: Rapid change of temperature |
| (17) | IEC 60512-11-9 Ed. 1.0:2002 (IEC 512-5-11i) | Climatic tests - Test 11i: Dry heat |
| (18) | IEC 60512-11-10 Ed. 1.0:2002 (IEC 512-6-11j) | Climatic tests - Test 11j: Cold |
| (19) | IEC 60512-11-12 Ed. 1.0:2002 (IEC 512-6-11m) | Climatic tests - Test 11m: Damp heat, cyclic |
| (20) | JEIDA 38 | Hydrogen sulfide test |
| (21) | MIL STD 883, Method 2016 | Physical dimensions |
| (22) | MIL STD 883, Method 1009 | Salt atmosphere corrosion test |

Appendix B (Normative) : Special Terms

The special terms that are used in this document should be defined and explained here. The terms should follow the Naming Convention.

B.1 Terminology

contact-pads side	The side with the card label. The TOP side of the card
non contact-pads side	The BOTTOM side of the card
Damp heat	Condition where temperature and humidity are controlled.
Discontinuity	Mechanical disconnection when a card is inserted in a connector
Keep Out Area	Where shall not print marks.
Micro-interrupt	Signal disconnection when a card is inserted in a connector
Via Hole Keep Out Zone	Where shall not place via hole.
warpage	Card condition out of flatness

B.2 Abbreviations

ANT1,ANT2	Antenna pins
CLK	Clock Input Pin
CMD	Command Pin
DAT	Data Pin
ESC	External Signal Contacts. Contact pads of a card.
ESD	Electrical Static Discharge
MSTG	microSD Task Group

Appendix C (Informative) : Card Detection Switch

This chapter describes how to use the card detection switch on the host connector. The SD Card detection switch is used to detect whether the SD is inserted to the host connector or not.

The host manufactures can choose either following a) or b) design method by host system requirements such as to achieve low power consumption, etc.

There are two types of the SD card detection switch implementations on the host connector.

a) **Normally Open**

The Card Detection Switch is open, while the SD Card is removed.

b) **Normally Closed**

The Card Detection Switch is closed while the SD Card is removed.

Table C - 1 shows the card detection switch types and its status:.

Card Detection Switch Types	SD Card is Removed	SD Card is Inserted
Normally open	OFF (open)	ON (closed)
Normally closed	ON (closed)	OFF (open)

Table C - 1 : Card Detection Switch Type and Status

The card insertion and removal sequence of SD card should take the following procedures:

a) SD card insertion sequence

Normally open type: The card detection switch should be turned on after all SD card contact pads are connected to the host connector contact pads.

Normally closed type: The card detection switch should be turned off after all SD card contact pads are connected to the host connector contact pads.

b) SD removal sequence

Normally open type: The card detection switch should be turned off when the SD card is just going to be removed and before any SD card contact pad is disconnected from the host connector contact pad.

Normally closed type: The card detection switch should be turned on when the SD is just going to be removed and before any SD contact pad is disconnected from the host connector contact pad.

The SD card detection switch with above insertion and removal sequence can be used for the SD host slot power control.

6.4.6 Example of Recommended Card Detection Switch Circuit

Figure C - 1 shows an example of recommended implementation of the card detection switch and related circuit.

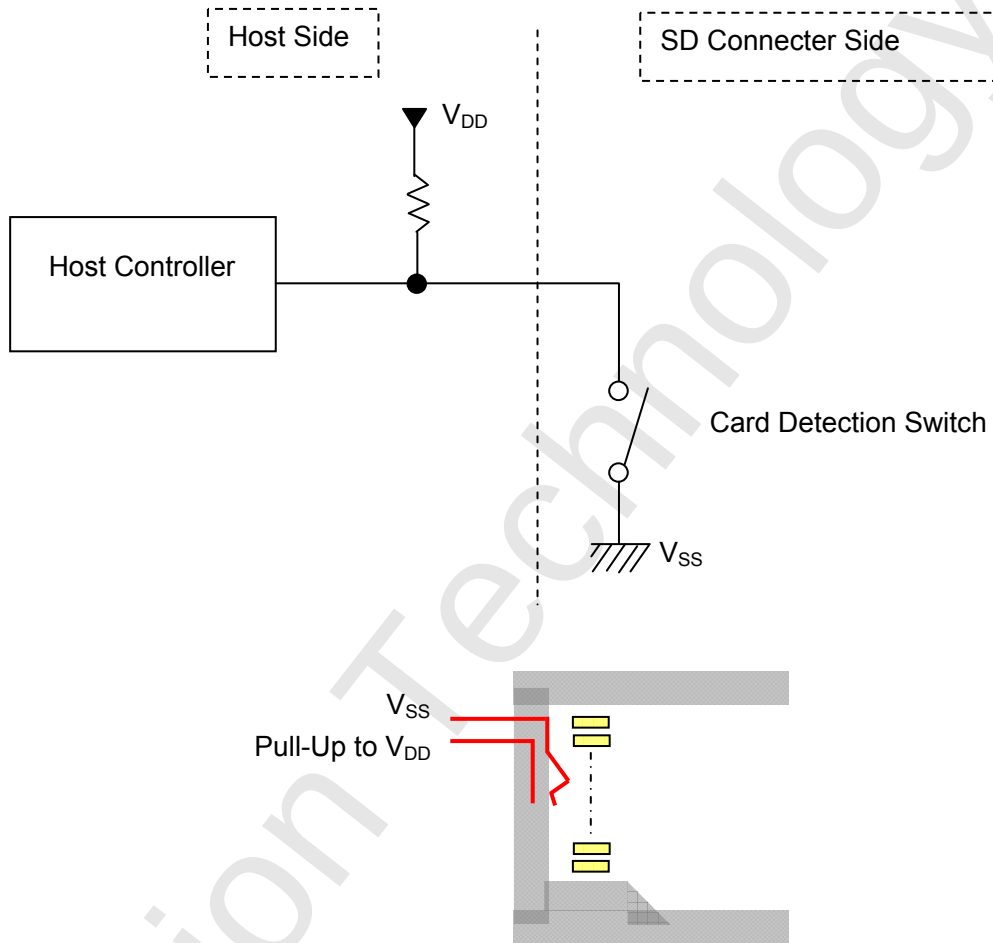


Figure C - 1 : Example of Recommended Card Detection Switch Circuit

6.4.7 Example of Not Recommended Card Detection Switch Circuit

Figure C - 2 shows an example of NOT recommended implementation of the SD card detection switch and related circuit. It is recommended that the SD card detection switch is not connected to V_{DD} directly for fail safe reasons. If the SD card detection switch is connected to V_{DD} , it may increase probability of V_{DD} - V_{SS} short circuit when some foreign substance is inserted into the host connector, or SD card is inserted while the host connector pin is broken, for some reason, and may cause the V_{DD} of the card detection switch line to accidentally contact the V_{SS} line.

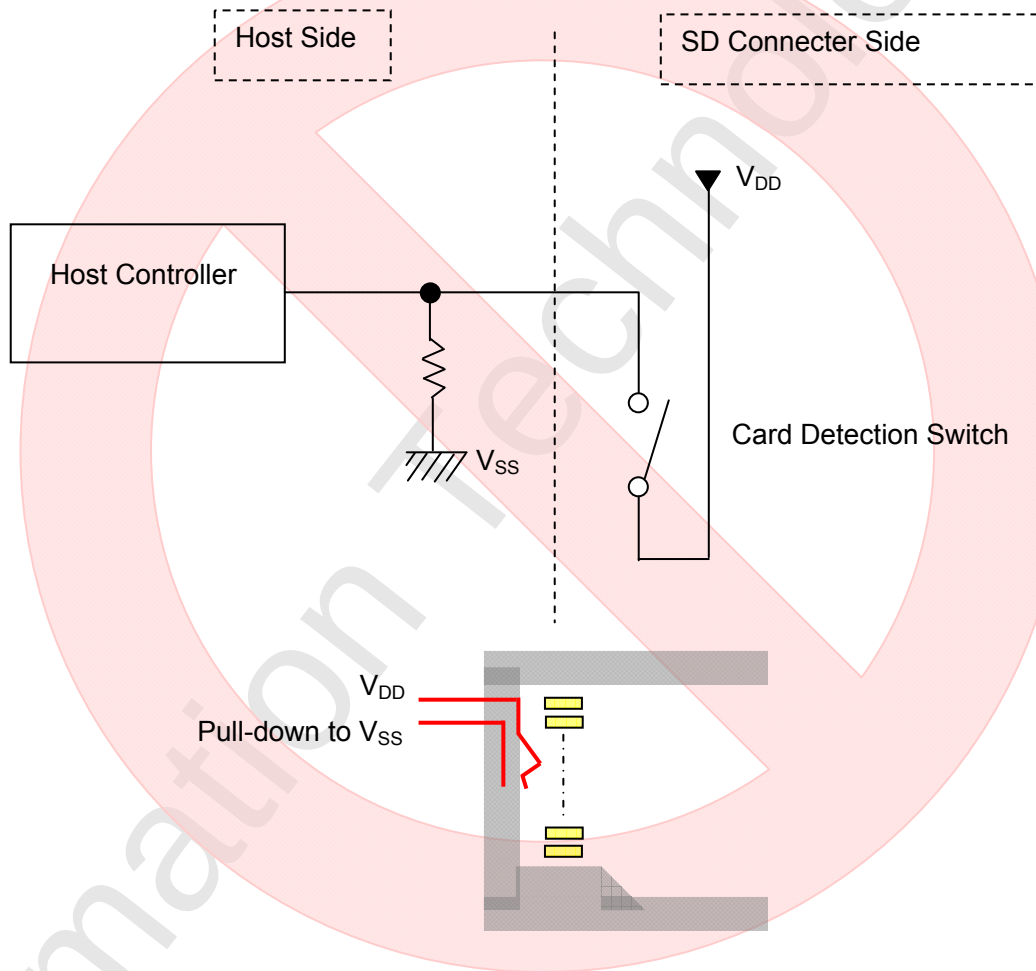


Figure C - 2 : Example of Not Recommended Card Detection Switch Circuit

Appendix D (Informative) : ESD Test Method

This section provides two ESD test methods that conform to the ESD requirements given in Section 3.1.3.

D.1 Contact Discharge Test

The following ESD test recommendation relates to the ESD requirement for contact pads 8.6.2 (1)(c) given in the Physical Layer Specification Version 3.01 (Section 6.8 is added by Supplementary Notes).

- 1) Testing Conditions
 - IEC 61000-4-2, contact discharge +/-2kV and +/-4kV150pF, 330Ohm
 - Five (5) times per voltage and polarity for each contact pad.
- 2) Test Procedure
 - Write Test Data into the SD Memory Card.
 - Set the discharge voltage to +2kV
 - Discharge to the pad (Five (5) times each pad, removing residual charge on the card between every contact discharge.)
 - Repeat with -2kV, +4kV and -4kV.
 - Check the Data and SD Memory Card Function.
- 3) Test Result
 - The SD Memory Card shall operate as specified.
 - The SD Memory Card shall retain the Data.

D.2 Air Discharge Test

The following ESD test recommendation relates to the ESD requirement for non contact pads 8.6.2 (2) given in the Physical Layer Specification Version 2.00 (Section 6.8 is added by Supplementary Notes).

- 1) Testing Conditions
 - IEC 61000-4-2, air discharge up to +/-15kV150pF, 330Ohm
- 2) Test Procedure
 - Write Test Data into the microSD Memory Card.
 - Mask the contact pads by insulating tape to avoid discharge (Figure D - 1).
 - set voltage for 4kV
 - Discharge toward the position (1) to (10) in order, each face. (Four (4) times top face as Figure D - 1 and five (5) times bottom face as Figure D - 2.) Remove residual charge on the card between each zap.
 - Repeat with -4kV, 8kV, -8kV, 15kV, -15kV.
 - Remove the insulating tape
 - Check the Data and microSD Memory Card Function.
- 3) Test Result
 - The SD Memory Card shall operate as specified.
 - The SD Memory Card shall retain the Data.

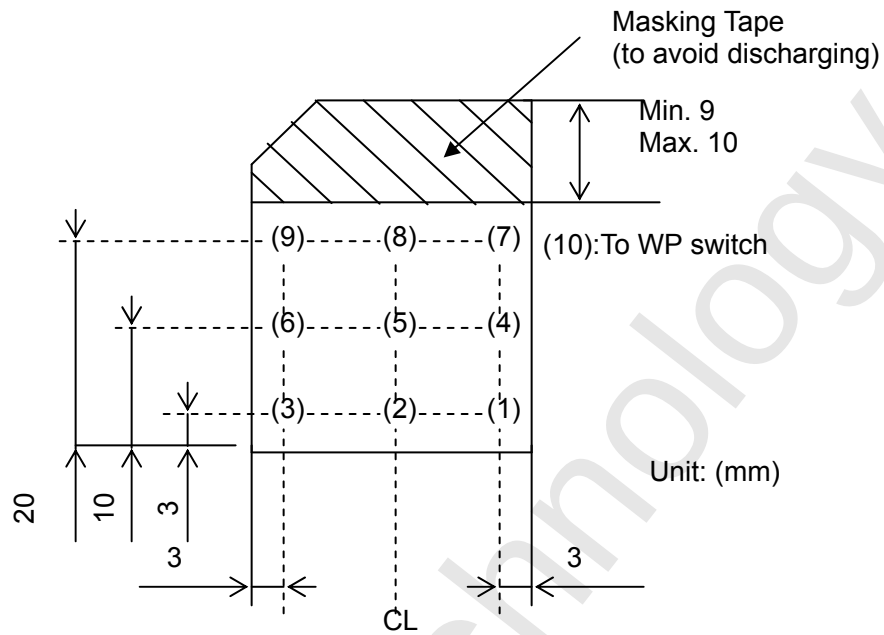


Figure D - 1 : Top Face Discharge Positions

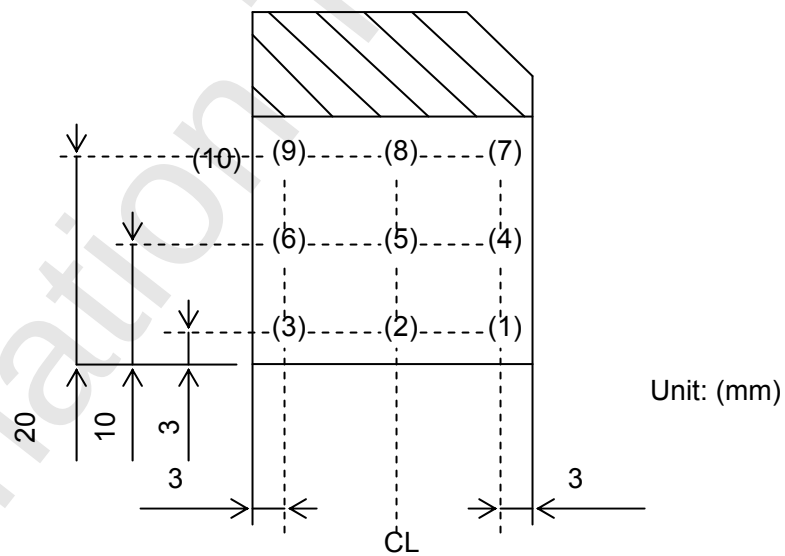


Figure D - 2 : Bottom Face Discharge

Appendix E (Informative) : Mechanical Testing Methods

This Appendix describes examples of test methods recommended by the SDA to be used for the compliance tests.

E.1 Bend Test Fixture Example

The Bend test relates to the Bending specification – given in Table 3-2, Table 5-2 and Table 6-2 Reliability and Durability.

Place the SD card on the fixture with surface facing upward.

Apply 10 N to the center of the surface of the SD card as shown in the figure below.

Time = 60 Sec.

Testing Fixture and Push Probe widths are more than 24 (SD Card width).

Card Test for Both sides Probe in the center 60- second.

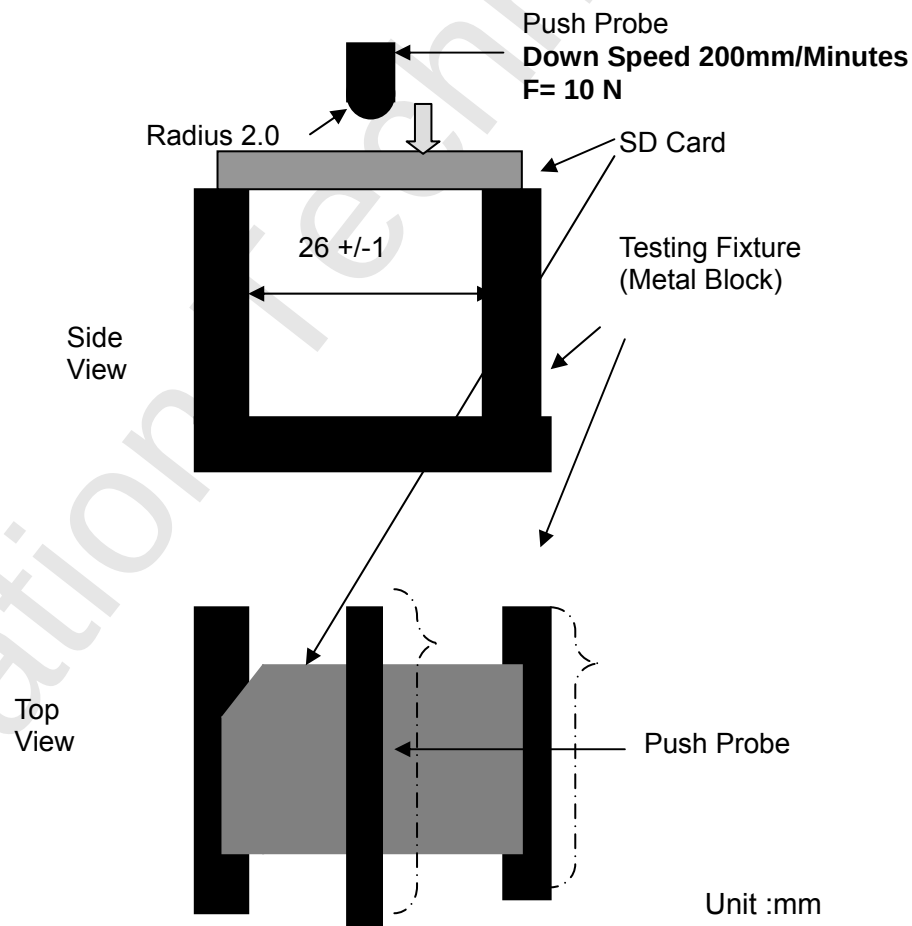


Figure E - 1 : Bending Test Fixture

E.2 Torque Test Fixture Example

The Torque test relates to the Torque specification – given in Table 3-2, Table 5-2 and Table 6-2 Reliability and Durability.

Apply torque to the unfixed fixture of the SD card.

Torque = 0.15 Nm.

Time= 30 Sec.

The maximum angle displacement is +/- 2.5 degrees on measurement.

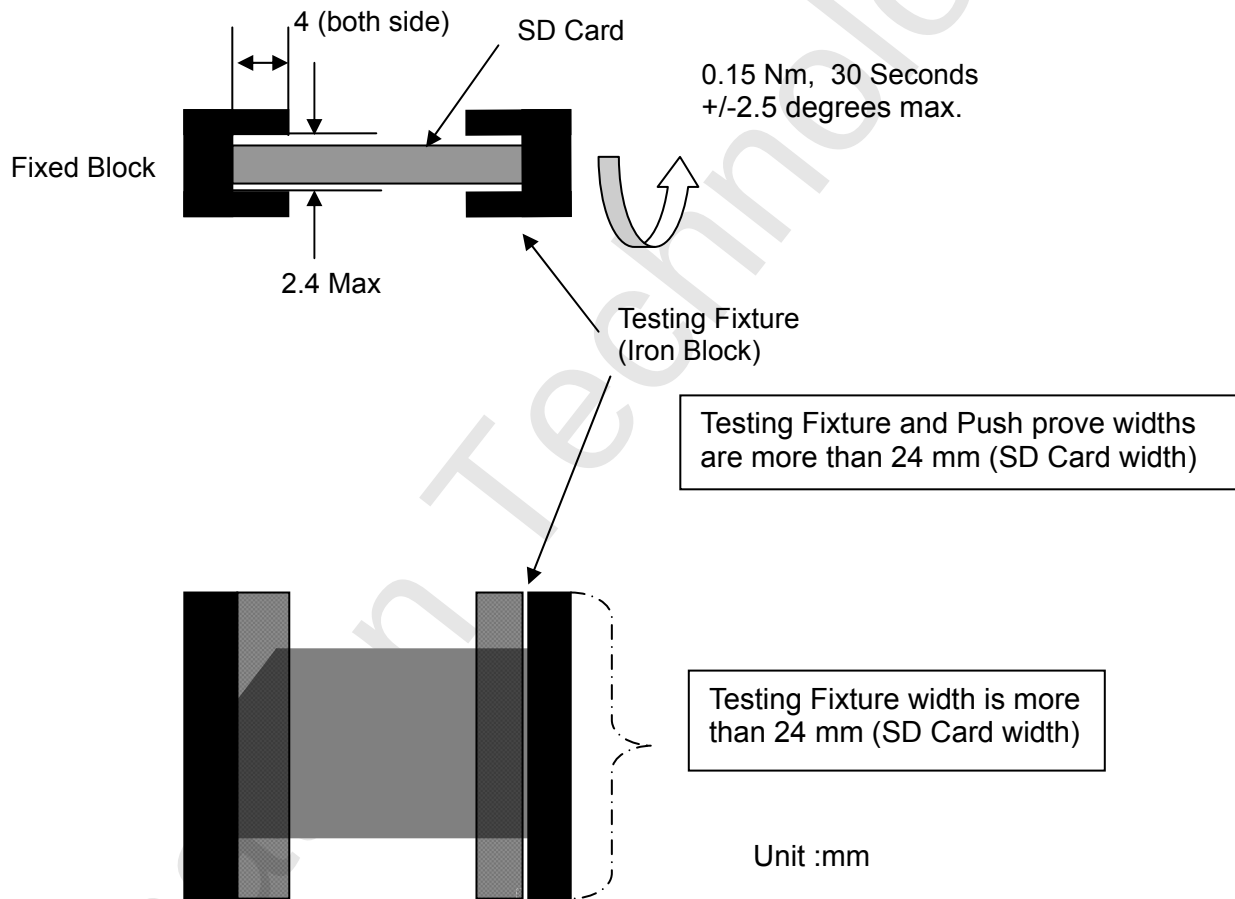


Figure E - 2 : Torque Test Fixture Example

E.3 Card Warpage Testing Fixture Example

The Warpage test relates to the Visual Inspection Shape and Form specification – given in Table 3-2, Table 5-2 and Table 6-2 Reliability and Durability.

The SD card should pass through the fixture by its weight.

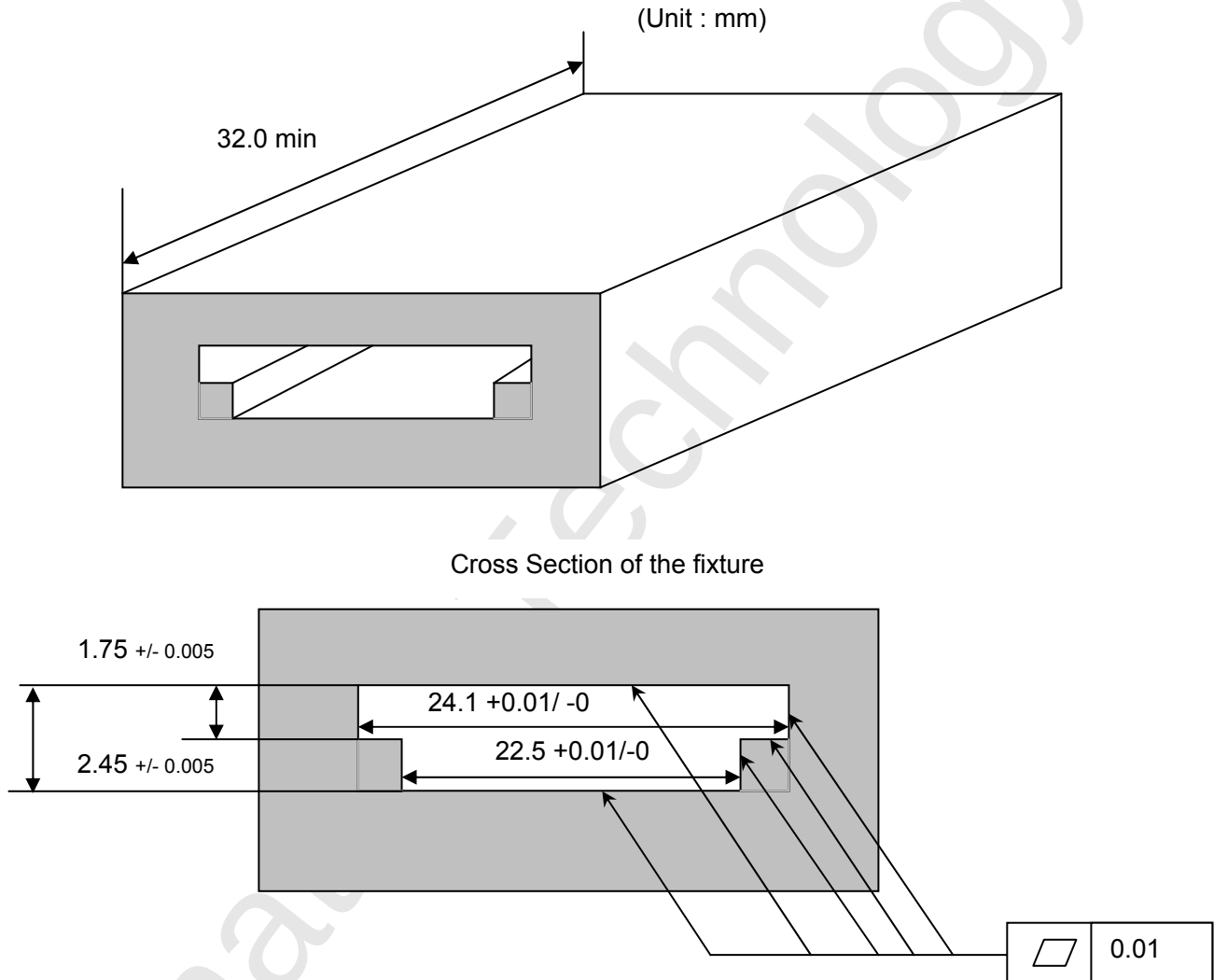


Figure E - 3 : Testing Fixture Example to Measure Card Warpage

E.4 Card Friction Test Fixture Example For Durability

The card friction test relates to the Visual Inspection Shape and Form specification – given in Table 3-2, Table 5-2 and Table 6-2 Reliability and Durability.

1) Test Condition

- Contact force of each pin in the Jig is $0.4 \pm 0.03\text{N}$.
- Extraction Speed is less than 1mm/minute.
- Condition: 20 to 25deg.C & 30% max. (Humidity)
- Each pin in the Jig is finished Gold plating on Ni plating.

2) Test Result

- The result of measurement by the jig mentioned in Figure E - 4 shall be less than 2.5N.

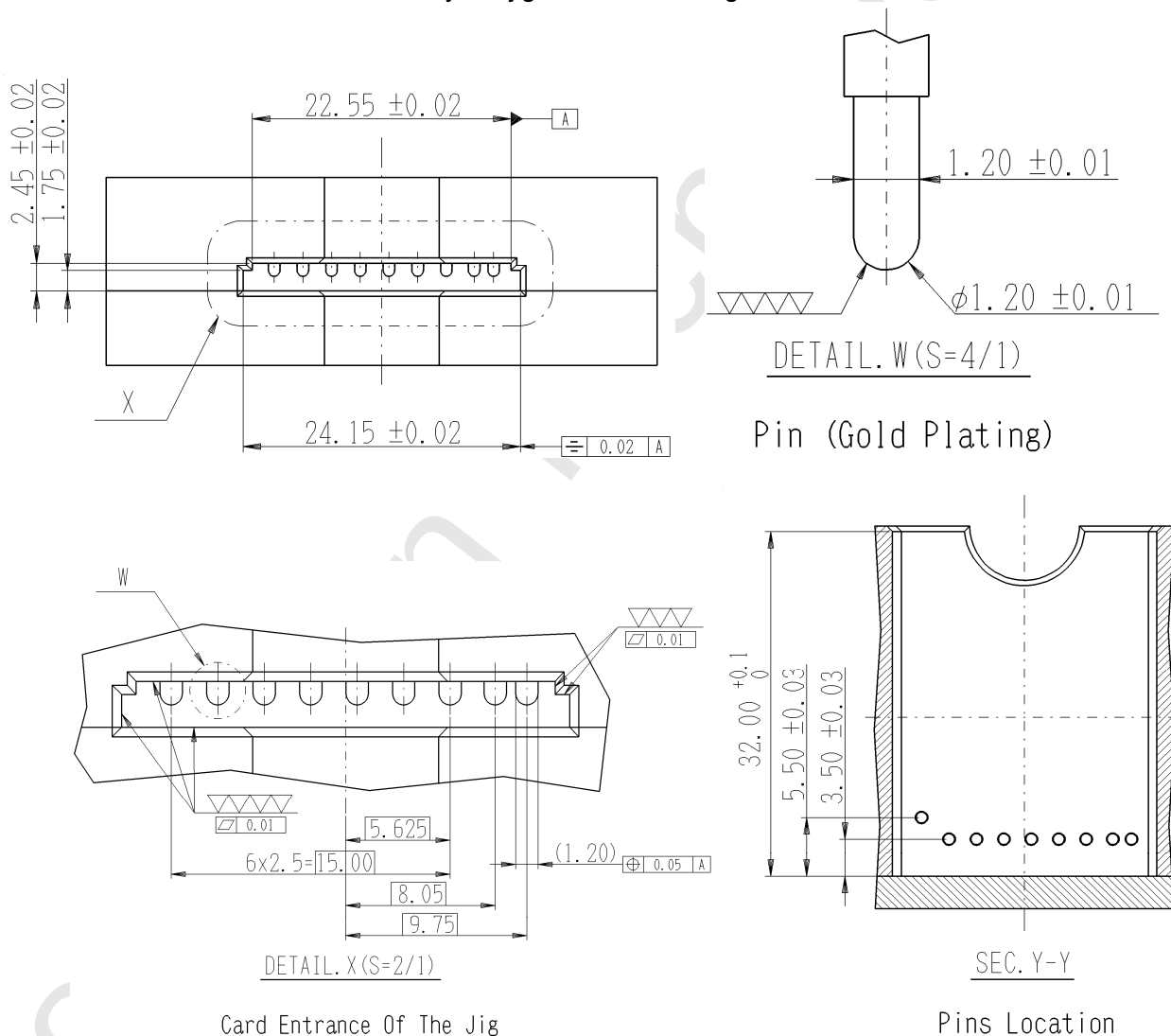


Figure E - 4 : Bottom Face Discharge Positions

E.5 WP-key Moving-Force Measured by Load Meter

The WP-key moving-force measurement test relates to the Minimum moving force of WP switch specification – given in Table 3-2, Table 5-2 and Table 6-2 Reliability and Durability.

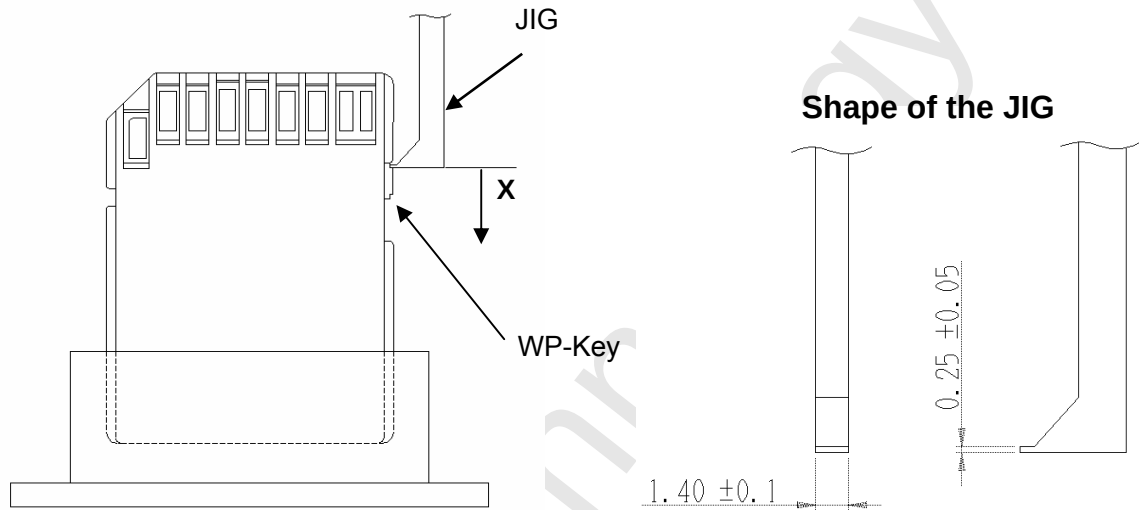


Figure E - 5 : Measurement Description

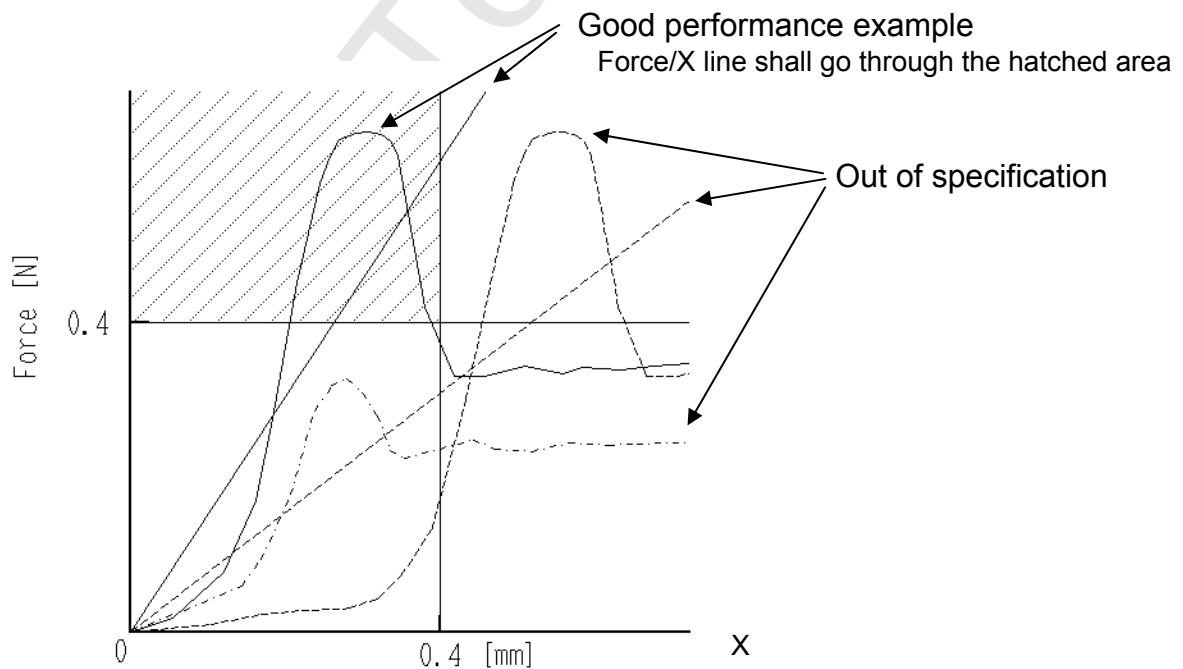


Figure E - 6 : Example of WP-key Moving-Force Measurement Results